Cascaded Spectral–Spatial Hyperspectral Anomaly Detection

Chein-I Chang, *Life Fellow, IEEE*, Chien-Yu Lin, Pau-Choo Chung, *Life Fellow, IEEE*, and Peter Fuming Hu

***Abstract*—An iterative spectral–spatial hyperspectral anomaly detection (ISSHAD) was recently developed to augment the data cube to be processed to construct a new data cube via a feedback process in an iterative manner. As a result of such a feedback process, hyperspectral anomaly detection (ISSHAD) significantly improves hyperspectral anomaly detection (HAD) performance. As an opposite approach to ISSHAD, this article develops a new approach to HAD, called cascaded spectral–spatial HAD (CSSHAD), which performs HAD by a cascade of anomaly detection modules (ADMs) in a feedforward network in which each cascaded ADM implements the same anomaly detector to produce an anomaly detection map (ADMap). Then, a spatial filter (SF) is designed to capture spatial information from the ADMap to generate a real-valued spatial filtered ADMap (SFMap) which will be used as the input to the next following ADM. Finally, the depth of CSSHAD is determined by an automatic stopping rule. To further improve anomaly detectability (AD) and background (BKG) suppressibility, a recently developed concept, effective anomaly space (EAS) is incorporated into CSSHAD to derive EAS-CSSHAD. Unlike ISSHAD which is a feedback HAD network, CSSHAD is a feedforward HAD network that operates a cascade of ADMs by feeding forward SFMaps in a similar manner that ISSHAD uses an iterative feedback process. Accordingly, CSSHAD and ISSHAD can be considered to be dual HAD methods as companion methods. Experimental results also demonstrate that CSSHAD performs comparably to ISSHAD with less computational complexity due to its use of a feedforward structure rather than a feedback process.**

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***Index Terms*—Anomaly detection (AD), anomaly-BKG separability (ABS), background (BKG), cascaded spectral–spatial hyperspectral anomaly detection (CSSHAD), effective anomaly space (EAS).**

NOMENCLATURE

Acronyms	Full Names
ABS	Anomaly-BKG separability.
AD	Anomaly detectability.
ADBS	Anomaly detection BKG suppressibility.
ADM	Anomaly detection module.
ADP	Anomaly detection probability.
Auto-AD	Autonomous hyperspectral anomaly detection.
BDP	BKG detection probability.
BKG	Background.
BS	BKG suppressibility.
EAS	Effective anomaly space.
ISSHAD	Iterative spectral–spatial hyperspectral anomaly detection.
ISSMax	Iterative spectral–spatial maximum.
JAD	Joint anomaly detection.
JBS	Joint BKG suppression.
LSDM-MoGs	Low-rank and sparse decomposition model with a mixture of Gaussians.
LRR	Low-rank representation.
OADP	Overall anomaly detection power.
PTA	Prior-based tensor approximation.
RGAE	Robust graph autoencoder.
ROC	Receiver operating characteristic.
RXAD	Reed–Xiaoli anomaly detector.
SFMap	Spatial filtered ADMap.

I. INTRODUCTION

FINDING anomalies is one of the fundamental tasks in hyperspectral imaging (HSI) and is also a great challenge due to the unavailability of target information [1], [2]. As a consequence, anomalies can appear in different forms. Some of them are of major interest, such as man-made objects and endmembers. Some others are of no interest, such as interferers and natural objects. According to the general consensus, spatial anomalies are objects which have distinct spatial characteristics different from their surrounding neighbor samples. Such spatial anomalies generally occur in video signal processing, such as moving targets frame by frame or dim small

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targets in infrared data processing. On the other hand, there are also spectral anomalies which are different from spatial anomalies in the sense that the former have spectrally distinct signatures from that of their neighboring samples. Of particular interest are subpixels or endmembers, which are generally cannot be visualized or identified by human vision or resolved by multispectral imaging (MSI). The issue between these two types of anomalies was briefly discussed in Chang et al. [3]. Nevertheless, there are also no clear-cut definitions of these two types of anomalies since many anomalies can be actually be specified spectrally as well as spatially, referred to as spectral–spatial anomalies. For example, mixed samples, small targets which only occupy a few pixels in the data. However, a common understanding of anomalies can be characterized by the following properties.

First of all, anomalies generally cannot be identified by prior knowledge or can be only barely visualized by human naked eyes. This property makes training sampling-based anomaly detectors difficult to work well. Second, the occurrence probability of anomalies is generally very low and unexpected. Third, once anomalies are present in the data, their populations are relatively small and cannot constitute a Gaussian distribution. Last but not least, anomalies are generally embedded BKG and corrupted by noise.

With all things considered above, several issues arising in hyperspectral anomaly detection (HAD) must be addressed. Most notable are how to improve anomaly-background separability (ABS), how to enhance anomaly detectability (AD), and how to increase background suppressibility (BS). To address these issues, Chang [4] recently coined a new concept, called EAS, and further proposed a sandwiched model which characterized anomalies being sandwiched between background (BKG) and noise. To bring out anomalies out from BKG to light, EAS assumes that BKG is described by the second order of data statistics and anomalies are specified by high orders of data statistics. Then, BKG can be removed by a data sphering using independent component analysis (ICA) [5] to retain anomalies in the ICA-generated independent component (IC) space. In order to avoid anomalies being mistaken for noise in the IC space, the concept of sparsity cardinality (SC) developed by go decomposition (GoDec) [6] is further used to eliminate noise. Since noise cannot be sparse, anomalies can be described as sparse signals by appropriately determine the value of SC by a sparsity order, k, with SC $\leq k$. By virtue of ICA and SC, an EAS can be, therefore, constructed for the sandwich model to effectively retain anomalies right between BKG and noise [4], [7], [8], [9], [10], [11].

A. Motivations

In recent years, deep learning (DL) has received considerable interest in various applications in HSI, particularly in HAD [12], where DL has advantages in extracting a large number of nonlinear features compared to traditional machine learning (ML) methods. Most DL-based works on HAD have been directed to how to estimate or reconstruct BKG that can be separated from anomalies, specifically those using the convolutional neural network (CNN) to produce the residual space which is supposed to retain anomalies. For example, Zhang et al. [13] developed a light CNN based on residual learning and BKG estimation to obtain the pseudo-BKG which can be subtracted from a hyperspectral image to obtain the anomaly matrix. Imani [14] proposed a collaborative representation for the BKG estimate and then used two distances obtained for inner and guard windows to calculate their difference to obtain anomaly scores. Zhao et al. [15] developed a hybrid transformer-CNN network for HAD to combine CNNs with transformers, where the CNN focuses on local modeling and the transformer addresses long-range modeling to ensure the accurate reconstruction of BKG by capturing both local and long-range dependencies. As an alternative to CNN, an autoencoder (AE) is also widely used to reconstruct BKG for HAD. Wang et al. [16] proposed an autonomous HAD (Auto-AD) network to reconstruct BKG, and anomalies are supposed to cause reconstruction errors. Since the number of anomalies is relatively small compared to that of BKG samples and anomalies also have a low occurrence probability in the data, an adaptive-weighted loss function is designed to further improve the anomaly reconstruction where the weights of potential anomalous pixels with large reconstruction errors are reduced during training via an adaptive-weighted loss function which assigns greater weights to potential anomalous pixels so as to increase the contrast of the anomalies against BKG. However, AE is sensitive to noise as well as anomalies during training and the relationships between pixels are also discarded. In order to tackle these problems, Fan et al. [17] proposed a robust graph AE (RGAE) for HAD, which first redesigned the objective function to make the network more robust to noise and anomalies. Then, a superpixel segmentation-based graph regularization term is incorporated into AE to preserve the geometric structure and spatial information simultaneously.

All the DL-based HAD methods share the same common network architecture, i.e., a feedforward multilayer neural network which requires training samples explicitly by *a priori* knowledge or implicitly by a *posteriori* training samples obtained from the data. The effectiveness of their detection performances is eventually determined by two key factors, training samples and models to be used to train the network, including an appropriate determination of its depth. Unfortunately, selecting training samples and tuning model parameters are empirical due to the nonexplainability of the network. In other words, a network may work well on some occasions but may not work with slight changes in training samples or parameters on other occasions. This leads to why so many DL-based HAD methods are proposed without specific guidelines.

B. Proposed Approach

Since most DL-based HAD methods take advantage of either the structures of CNN or AE, they still suffer from the same issues in selecting training samples and fine-tuning model parameters. So, this article looks into the same feedforward network from a completely different perspective.

The proposed idea is inspired by two recent works which look seemingly opposite approaches in terms of network structures, but they indeed shed light on the same idea in how to utilize spatial filters (SFs) to capture spatial information from the processed data to improve system performance. One is the multispatial filtering module cascaded system (MSFMCS)

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developed for hyperspectral image classification, which is a cascaded feedforward system [18]. It used a set of SFs to obtain classification information from each of the classification maps in one module and fed forward such information to the following concatenated module. The other is the ISSHAD developed for HAD, which is a feedback network system [9]. It also used an SF to obtain spatial information from an ADMap in an iteration manner and then fed the SFMap back to be added to augment the current data cube as a new data cube used for the next iteration. By realizing such striking similarity, this article combines the ideas of MSFMCS and ISSHAD to develop a new approach which designs a cascade of ADMs, called cascaded spectral–spatial HAD (CSSHAD), where one ADM is concatenated with another ADM in sequence. Each ADM includes an SFMap captured by CSSHAD to perform HAD and then feeds the resulting SFMap forward to the next following ADM in a similar manner that MSFMCS does. So, technically speaking, CSSHAD can be considered as a companion version of MSFMCS for HAD. On the other hand, CSSHAD can be considered as an counterepart feedforward version of ISSHAD. There are two great benefits that can be derived from CSSHAD. First of all, no model parameters are required. Second of all, no training samples are needed to train the network. Both address the issues encountered in DL-based HAD methods in Section I-A.

As also shown in [4], the HAD performance can be further improved by operating an anomaly detector in an EAS. Implementing CSSHAD in EAS yields EAS-CSSHAD which can significantly improve ABS, AD, and BS. It should be noted that such EAS-CSSHAD does not depend on any specific anomaly detector. Therefore, CSSHAD and EAS-CSSHAD can be used as a general platform for any arbitrary anomaly detector to improve its HAD performance.

C. Contributions

Several novel contributions are summarized as follows.

1) A new cascaded feedforward network system is developed for HAD, CSSHAD which consists of a cascade of ADMs, one concatenated with another in a feedforward manner. Specifically, no weights are connected between two concatenated ADMs. Therefore, no training is needed for CSSHAD.
2) Unlike DL-based methods, CSSHAD does not require model parameters and training samples to train the network. Most importantly, the depth of CSSHAD can be determined by an automatic stopping rule.
3) To improve CSSHAD in AD and BS, an EAS-CSSHAD is further developed to implement CSSHAD in EAS. Since CSSHAD does not use feedback information to augment the current data cube to a new data cube as ISSHAD does, it reduces the computation complexity of ISSHAD, while still retaining its HAD performance.
4) Both CSSHAD and EAS-CSSHAD do not specify a particular anomaly detector. As a consequence, CSSHAD provides a general platform for any given anomaly detector to improve its performance in anomaly detection.

The remainder of this article is organized as follows. Section II briefly reviews related works. Section III presents a cascaded structure for HAD, CSSHAD, along with its extension to EAS, EAS-CSSHAD. In particular, a comparative study between CSSHAD and ISSHAD is conducted. Section IV introduces the real hyperspectral image scenes used in the experimental sections. Section V presents experimental results and discussions. Section VI further discusses CSSHAD with interesting findings. Section VII concludes the manuscript.

II. Related Works

Many works on HAD have been investigated and reported in the literature. An early approach was developed by Reed and Yu [19], referred to as the RXAD which was derived from the generalized likelihood ratio test (GLRT) and turned out to be the Mahalanobis distance [2], [20]. Since then many its variants have been developed, such as local RXAD [21], [22], support vector data description (SVDD) [23], and techniques weighting-RXAD and linear filters [24]. In addition, more complex models like the dual window-based eigen separation transform (DWEST) [25], collaborative representation-based anomaly detector (CRD) [26], multiple-window AD (MWAD) [27], and kernel-based RXAD [28]. In particular, RXAD was extended to by replacing the global sample auto-covariance matrix $\mathbf{K}$ with the global sample autocorrelation matrix $\mathbf{R}$ [29] and later proved mathematically by Chang [30].

Recent advancements have been focused on addressing a critical issue arising in HAD, BS, as noted in Section I-A. In addition to DL-based methods [13], [14], [15] and AE-based methods [16], [17] reviewed above, there are also other methods widely used and developed for HAD. For example, low-rank and sparse representation-based methods have gained prominence, such as LRR [31]. These methods were developed to represent BKG as a low-rank space while treating anomalies as sparse signals. Techniques like DL networks [32], [33], AEs [17], [33], and generative adversarial nets (GANs) [34], [35], [36].

In terms of enhancing AD, various strategies have been investigated such as Gaussian mixture models (GMMs) [37], robust nonlinear AD [38], sparsity score estimation [39], and BKG-anomaly component projection and separation optimized (BASO) filter [40]. These approaches focus on either BKG estimation/reconstruction or anomaly detection enhancement, with some recent models attempting to address both simultaneously. One such dual-focused approach is the low-rank and sparse matrix decomposition (LRaSMD) model, which has been further explored through robust principal component analysis (RPCA) [41] and its variants, including robust subspace learning (RSL) [42], [43], [44], [45] and GoDec [46]. These models aim to separating BKG and anomalies into distinct spaces, enhancing anomaly detection. As an alternative model to LRaSMD, CDA decomposes a data matrix into the principal component (PC) space and IC space [7].

Despite the above-mentioned works, the challenge of how to cope with BS remains. To address this issue, the Auto-AD network [16] and the LSDM-MoGs [47], [48] were developed to reconstruct BKG more effectively to achieve better BS. In addition, PTA [49] took advantage of Tucker tensor decomposition to separate the data space into a BKG tensor and an anomaly tensor so as to enhance HAD performance. EAS

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in [4] was constructed via a sandwich model by considering anomalies to be embedded between BKG and noise. In this case, EAS can remove BKG and noise to enhance AD. In the meantime, it can also better suppress BKG. All these developments reflect a significant evolution in HAD, offering better and more effective solutions to HAD.

III. CSSHAD AND EAS-CSSHAD

CSSHAD has a similar network to a feedforward neural network with several key different components. First of all, unlike the traditional feedforward multilayer perceptron with all layers fully connected, CSSHAD consists of a cascade of ADMs in a feedforward manner, one module concatenated to another successively, where each ADM implements an independent anomaly detector. Second, two successive ADMs are concatented by an input–output relationship and there are no weights connected between them. Third, no training samples are required to train the network. Fourthly, each ADM in CSSHAD stands alone as an independent module. Finally, the number of modules defined as the depth of CSSHAD can be determined by an automatic stopping rule. This is quite different from many DL-based methods where the depth is determined empirically *a priori*.

More specifically, in CSSHAD, each ADM between the initial input and final output ADMs employs the same anomaly detector to produce an ADMap and then utilizes an SF to capture spatial information of the ADMap to generate a real-valued SFMap which will be used as the input to the next following ADM. The depth of CSSHAD is then determined by the Tamimoto index (TI) [50], which calculates the overlapped area of two binary maps (BMaps) obtained by thresholding the SFMaps in two concatenated ADMs using Otsu's method [51] to see if their overlapped area is greater than a prescribed value. If yes, CSSHAD is terminated. Otherwise, CSSHAD continues.

A. Detailed Implementation of CSSHAD

The initial module of CSSHAD started by taking an original hyperspectral data cube Ω^0 and applying an anomaly detector to produce an ADMap^0. Next, Otsu's thresholding method is applied to ADMap^0 to generate a BMap^0 from which a foreground map FGMap^0 can be produced from BMap^0. Then, FGMap^0 was combined with the original data cube Ω^0 to augment data cube $\mathbf{X^0}$, which is then input to the first ADM. In the jth ADM (for $j \geq 1$) in the cascade, the process is repeated over again with the augmented data cube $\mathbf{X}^{j-1}$ input from the $(j-1)$st ADM. Then, the same anomaly detector is implemented in the jth ADM to generate ADMap^j. Then, Otsu's method is applied to binarize ADMap^j to produce BMap^j with a one indicating an anomalous pixel and a zero representing a BKG sample. A stopping rule calculates TI to determine if the process should be terminated. If TI is greater than a prescribed threshold value, CSSHAD is terminated. Otherwise, the process is continued. In this case, it finds $\text{maxBMap}^j = \max\{\text{BMap}^j, \text{BMap}^{j-1}\}$ and then multiplies maxBMap^j with ADMap^j elementwise to produce a foreground map FGMap^j. Finally, the original data cube Ω^0, FGMap^j is combined the original data cube Ω^0 to form a new data cube $\mathbf{X}^j$ used as input to the $(j+1)$st ADM. The entire process is repeated over again by setting $j+1 \leftarrow j$ until a stopping rule is satisfied.

A detailed step-by-step implementation of CSSHAD is summarized in the following algorithm.

It should be noted that the bounds in error reduction are actually dictated by the values of TI in (3) obtained in each ADM implemented in the CSSHAD.

B. Discussions on Core Concepts of CSSHAD

The core concept of CSSHAD is the introduction of a network which is made up of a series of ADMs cascaded one after another in a feedforward manner. Each ADM uses an SF to capture spatial information from an ADMap to obtain an SFMap as the output which is fed forward as the input to next ADM. As a result of including more spatial information about the ADMaps in ADMs, the anomaly detectability and separability are then increased and enhanced ADM-by-ADM in each feedforward step. As for EAS, it was shown in [4] in great detail on how EAS can enhance anomaly detectability. Since the details are provided in [4], there is no need to include its details in this article.

As noted, CSSHAD requires no training samples nor fine-tuned model parameters to train the network. As a consequence, there is no sensitivity issue involved in CSSHAD. As for datasets used for anomaly detection, this issue was addressed in [3]. As a matter of fact, many datasets used for HAD reported in the literature are indeed not applicable to HAD. Specifically, San Diego airport data scene has been explored in great details and investigated extensively in [3] and [4]. Despite that, many datasets have been used for HAD, to the authors' best knowledge, their ground truths (GTs) are obtained based on visual inspection without prior knowledge. In this case, many real invisible anomalies which are actually present in the datasets but cannot be identified by the human naked eyes. So, using appropriate datasets for HAD is more important than using many datasets for experiments, such as San Diego airport, Gulfport, Texas coast, and Los Angeles, which are not suitable for HAD as demonstrated in [3].

It should be noted that CSSHAD is independent of its used base anomaly detector. It provides a general platform to improve and enhance AD and BS for any arbitrary anomaly detector. So, the limitation on base anomaly detector, scalability to large HSIs, noise robustness, and real-world applicability are all determined by the anomaly detector used in CSSHAD not CSSHAD.

C. Comparison Between ISSHAD and CSSHAD

Despite the fact that CSSHAD and ISSHAD are developed from different design rationales, it is worthwhile exploring their dissimilarities and similarities.

1) Dissimilarities Between CSSHAD and ISSHAD: There are two key differences between CSSHAD and ISSHAD as follows.

a) *Network structure:*

i) CSSHAD: Feedforward network with depth determined by the number of ADMs, n_{ADM}.
According to [52], CSSHAD can be considered as a moving average (MA) model.

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ii) ISSHAD: Feedback network with depth determined by the number of iterations n_{iter}.
According to [52], ISSHAD can be considered as an autoregressive (AR) model.

b) *New Data Cube Construction:* Assume that the original data cube is denoted by $\mathbf{X}$ and let $\Omega^0 = \mathbf{X}$.

i) CSSHAD

1) Initial ADM: $\mathrm{X}^0 = \Omega^0 \cup \mathrm{FGMap}^0$ obtained from (1) and (2).
2) jth ADM with $j \geq 1$: $\mathrm{X}^j = \Omega^0 \cup \mathrm{FGMap}^j$ obtained from (5) and (6).

The new data cube constructed in the jth ADM takes the union of the original data cube $\Omega^0 = \mathbf{X}$ with FGMap^j which is the output produced by the jth ADM.

ii) ISSHAD

1) Initial condition: $\Omega^1 = \Omega^0 \cup \mathrm{FGMap}^0$ obtained from (1) and (2).
2) jth iteration with $j \geq 1$: $\Omega^j = \Omega^{j-1} \cup \mathrm{FGMap}^j$.

The new data cube constructed by the jth iteration takes the union of the input at the $(j-1)$st iteration, data cube Ω^j with FGMap^j which is the output produced by the jth ADM. More specifically, $\Omega^j = \Omega^0 \cup (\cup_{k=1}^{j} \mathrm{FGMap}^k)$.

As can be seen from the new data cube construction, the difference between CSSHAD and ISSHAD is that the output $\mathbf{X}^j$ of the jth ADM in CSSHAD unions $\Omega^0 = \mathbf{X}$ only with its FGMap^j, whereas the output Ω^j of the jth iteration in ISSHAD takes a union of Ω^{j-1} with FGMap^j where $\Omega^{j-1} = \Omega^0 \cup (\cup_{k=1}^{j-1} \mathrm{FGMap}^k)$. As a result, if the original data cube $\Omega^0 = \mathbf{X}$ has L bands, the $\mathbf{X}^j$ of the jth ADM in CSSHAD has only $L+1$ bands compared to the output of the jth ISSHAD, Ω^j, which has $L + j$ bands. It should be noted that the extra band in $\mathbf{X}^j$ is provided by FGMap^j which is the spatial information map obtained from ADMap^j produced by the jth ADM and the j bands in Ω^j produced by the jth iteration in ISSHAD are $\{\mathrm{FGMap}^k\}_{k=1}^{j}$ which are the spatial information map obtained from all the FGMap^k at previous iterations or $1 \leq k \leq j$. It is the FGMap^j for CSSHAD and $\{\mathrm{FGMap}^k\}_{k=1}^{j}$, both of which can help CSSHAD and ISSHAD improve HAD.

2) Similarities Between CSSHAD and ISSHAD: There are also two key concepts shared by CSSHAD and ISSHAD.

i) Idea of *u*sing an SF to *g*enerate SFMaps. In order to improve HAD, an SF, such as a Gaussian filter, is applied to an ADMap to capture spatial information from the ADMap, SFMap, which is used to construct a new data cube for CSSHAD and ISSHAD.
ii) Stopping rule using TI defined by (3) to terminate CSSHAD and ISSHAD.

The criterion TI is also referred to as the Jaccard Index or intersection over union (IoU) for objection detection widely used in computer vision [53]. It measures the ratio of the intersection to the union of two sets. These two sets are BMaps produced by two concatenated ADMs in CSSHAD and two successive iterations in ISSHAD. As a summary, Table I lists dissimilarities and similarities between CSSHAD and ISSHAD described above.

Algorithm 1 For CSSHAD

The initial module:

1. 1st step: let HSI be the original data cube Ω^0. Implement an anomaly detector on Ω^0 to produce an anomaly detection map ADMap^0.
2. 2nd step: apply Otsu's method on ADMap^0 to create a binary map, BMap^0.
3. 3rd step: create a foreground map, FGMap^0 by
$$\mathrm{FGMap}^0_{m,n} = \mathrm{ADMap}^0_{m,n} \times \mathrm{BMap}^0_{m,n} \tag{1}$$
4. 4th step: form a data cube X^0 where
$$\mathrm{X}^0 = \Omega^0 \cup \mathrm{FGMap}^0 \tag{2}$$
which is also the input data cube for the 1st ADM. *The jth ADMap*
5. 5th step: implement an anomaly detector on X^{j-1} to produce an anomaly detection map ADMap^j.
6. 6th step: apply Otsu's method to ADMap^j to produce a binary map, BMap^j.
7. 7th step: check the following stopping rule
$$TI^j = \frac{|\mathrm{BMap}^j \cap \mathrm{BMap}^{j-1}|}{|\mathrm{BMap}^j \cup \mathrm{BMap}^{j-1}|} > \varepsilon \tag{3}$$
If (3) is true, then the process is terminated. Otherwise, continue.
8. 8th step: calculate $\mathrm{maxBMap}^j$ by
$$\mathrm{maxBMap}^j = \max\left\{\mathrm{BMap}^j, \mathrm{BMap}^{j-1}\right\} \tag{4}$$
9. 9th step: create a foreground map, FGMap^j, by
$$\mathrm{FGMap}^j_{m,n} = \mathrm{ADMap}^j_{m,n} \times \mathrm{maxBMap}^j_{m,n} \tag{5}$$
10. 10th step: form a data cube X^j where
$$\mathrm{X}^j = \Omega^0 \cup \mathrm{FGMap}^j \tag{6}$$
11. 11th step: Let $j + 1 \leftarrow j$, and repeat steps 5 to 10.

Algorithm 2 For EAS-CSSHAD

Let Ω^0 be the full band set
1. Construct EAS from Ω^0, denoted by Ω^{EAS}.
2. Implement CSSHAD on Ω^{EAS}.

D. EAS-CSSHAD

As mentioned in the introduction, anomalies are often sandwiched between BKG and noise, making HAD very challenging. According to [9], ISSHAD implemented in conjunction with EAS indeed improved HAD performance. By taking advantage of EAS, CSSHAD can be further improved by implementing in EAS to replace the original data space to derive a new variant of CSSHAD, called EAS-CSSHAD. Since the great details of the construction of EAS can be found in [4], [7], [8], [9], [10], and [11], their details are not included here but referred to these references.

A block diagram of details in implementing EAS-CSSHAD step-by-step is depicted in Fig. 1. It should be noted that CSSHAD is part of EAS-CSSHAD as described in

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TABLE I
DISSIMILARITIES AND SIMILARITIES OF CSSHAD AND ISSHAD

	dissimilarities	
	CSSHAD	ISSHAD
Network structure	feedforward	feedback
New data cube construction	$\mathbf{X}^j = \mathbf{\Omega}^0 \cup \text{FGMap}^j$ for the jth ADM and $\mathbf{X}^0$ = original data cube	$\mathbf{\Omega}^j = \mathbf{\Omega}^{j-1} \cup \left(\cup_{k=1}^{j} \text{FGMap}^k\right)$ for the jth iteration and $\mathbf{\Omega}^0 = \mathbf{X}$ = original data cube
Depth of network	number of ADMs, n_{ADM}	number of iterations, n_{iter}
Signal model [52]	Moving average (MA)	Autoregressive (AR)
	similarities	
	CSSHAD	ISSHAD
Using a spatial filter to capture spatial information from ADMaps	SFMaps	SFMaps
HAD	No specified anomaly detector	No specified anomaly detector
Improvement of HAD	EAS	EAS
Stopping rule	TI	TI

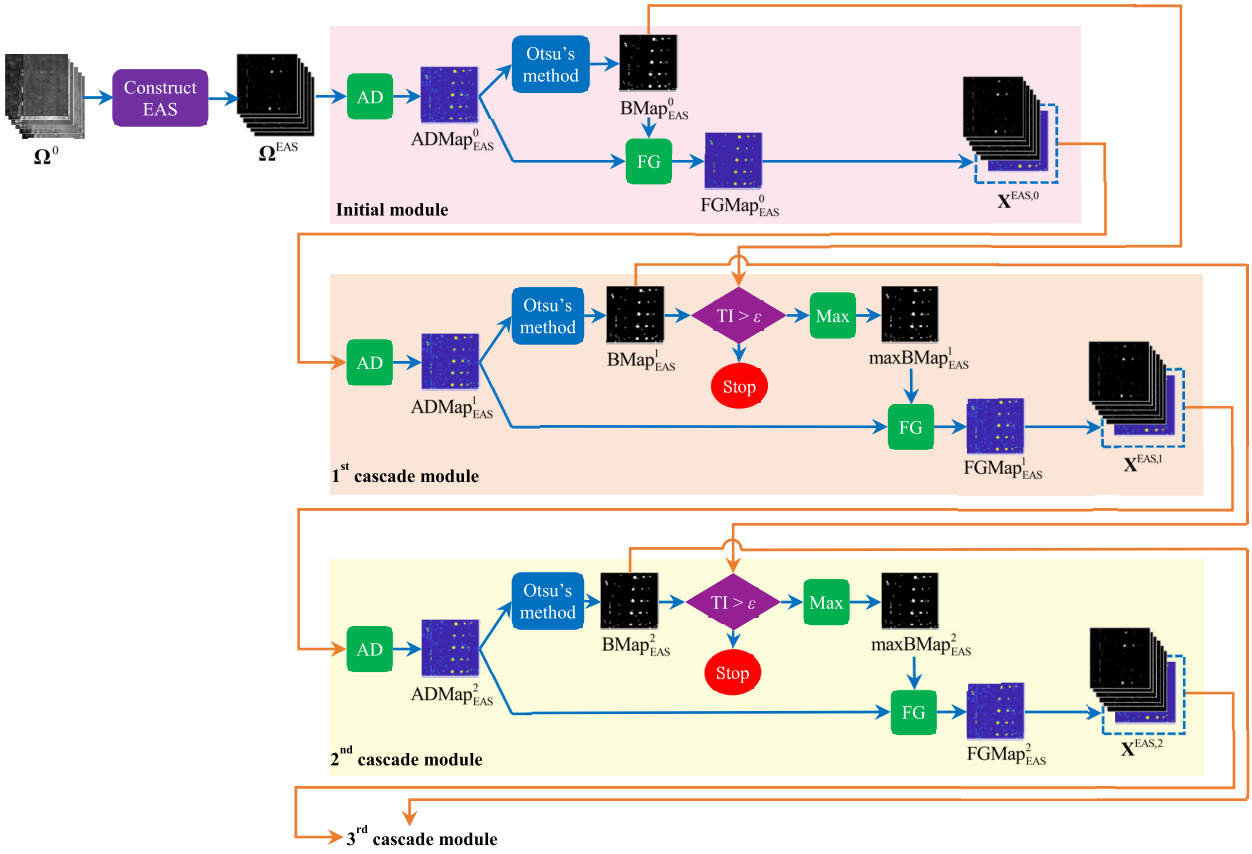


Fig. 1. Graphic block diagram of the initial module and first two cascade modules of EAS-CSSHAD.

Algorithm 2, there is no need of including its block diagram here.

IV. REAL IMAGES FOR EXPERIMENTS

In this section, we introduce four images to show the performance of the CSSHAD method across different datasets.

A. HYDICE 15-Panel Scene

Fig. 2(a) presents an airborne hyperspectral digital imagery collection experiment (HYDICE) scene captured in August 1995 from an altitude of 10 000 ft, with a ground sampling distance of about 1.56 m. Various studies have extensively explored this scene [1]. It comprises 169 bands used in experiments, with specific bands (1–3, 202–210, 101–112, and 137–153) removed due to low signal/high noise and water vapor absorption.

According to the GT, Fig. 2(b) shows 15 square panels of different sizes. Due to the 1.56-m ground sampling distance, each panel in the first column (except the first row) contains two pixels highlighted in red (p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5). The remaining 11 panels have one red panel pixel each. Fig. 2(b) precisely shows the spatial locations of these 19 red panel pixels, with yellow pixels indicating panel pixels mixed with the BKG.

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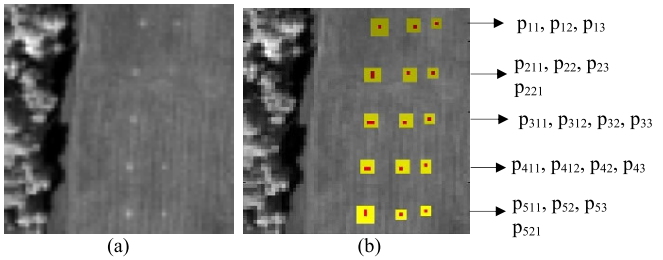


Fig. 2. (a) HYDICE15-panel scene which contains 15 panels and (b) GT map of spatial locations of 19 R panel pixels.

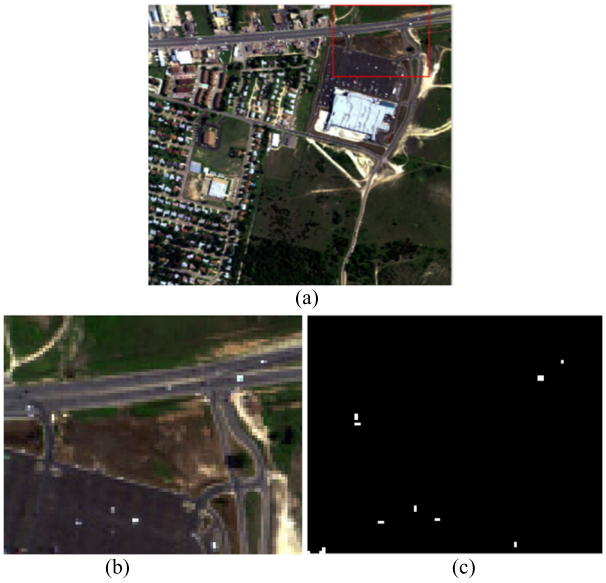


Fig. 3. HYDICE urban scene. (a) Entire image. (b) Pseudo-color selected area. (c) GT map.

It is worth noting that this panel scene is very valuable and was particularly designed for real-world application scenarios where the GT was collected by ground crews right at the scene. The materials made for panels were the same fabrics made for parachutes with different paints, particularly for search and rescue in sea or remote mountain areas. Three different panel sizes, 3, 2, and 1 m, were used for different purposes, that is, pure panel pixels in the 1st column for endmember finding/extraction, mixed panel pixels in the 2nd column for mixed pixel analysis, and subpixel panel pixels in the 3rd column for subpixel detection. Due to their relatively small sizes and spectral properties, all these 19 R-panel pixels in Fig. 2 can be considered as anomalies.

B. HYDICE Urban Scene

A second dataset features another HYDICE image presented in pseudo-color as shown in Fig. 3(a). The image is an urban scene consisting of 210 spectral bands, 174 of which were selected for experiments after removing noise and water absorption bands.

A region of size 80 × 100 pixels was chosen and cropped from the upper right corner of the scene shown in Fig. 3(b). Fig. 3(c) shows the corresponding GT map, where 21 pixels have been identified as anomalies: cars and roofs. The image can be downloaded from the Remote Sensing Laboratory website.[1]

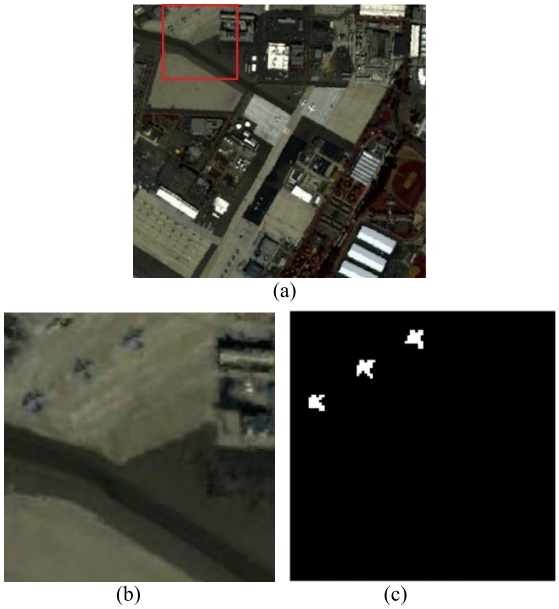


Fig. 4. San Diego airport scene. (a) Entire image. (b) Selected area. (c) GT map.

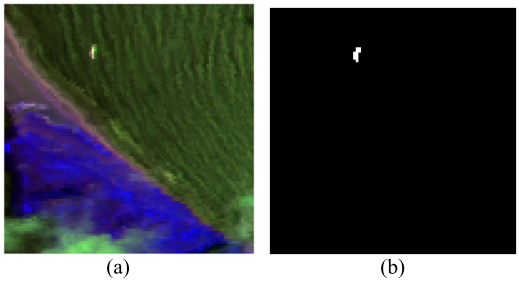


Fig. 5. AVIRIS Bay Champagne beach scene. (a) Pseudo-color scene. (b) GT map.

TABLE II
PARAMETERS SETTING FOR TEST AD METHODS

LRR [31]	$\beta = 0.1$
LDSM-MoG [47]	$(l_0,K) = (20,2)$ $(t_0,\mu_0,a_0,b_0,c_0,d_0,\alpha_0,\beta_0) = (10^{-3},0,10^{-6},10^{-6},10^{-6},10^{-6},10^{-6},10^{-6})$
PTA [49]	$(\gamma,\alpha,\mu,\beta,\tau,\text{maxIter}) = (1,1.7,0.069,0.1,400)$
RGAE [17]	$(\lambda,S,d) = (10^{-2},150,100)$
Auto-AD [16]	stopping threshold of MSE loss = 1×10^{-5}; number of layers = 5; learning rate = 1×10^{-2}; maxIter = 1001; dimensionality of the feature map = 128.
Max number to terminate CSSAD	20

C. San Diego Airport Scene

A third dataset is a scene of the San Diego airport in California, USA, shown in Fig. 4. It was captured by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) with main target features, roofs, shadows, and grass.

[1] https://rslab.ut.ac.ir/data

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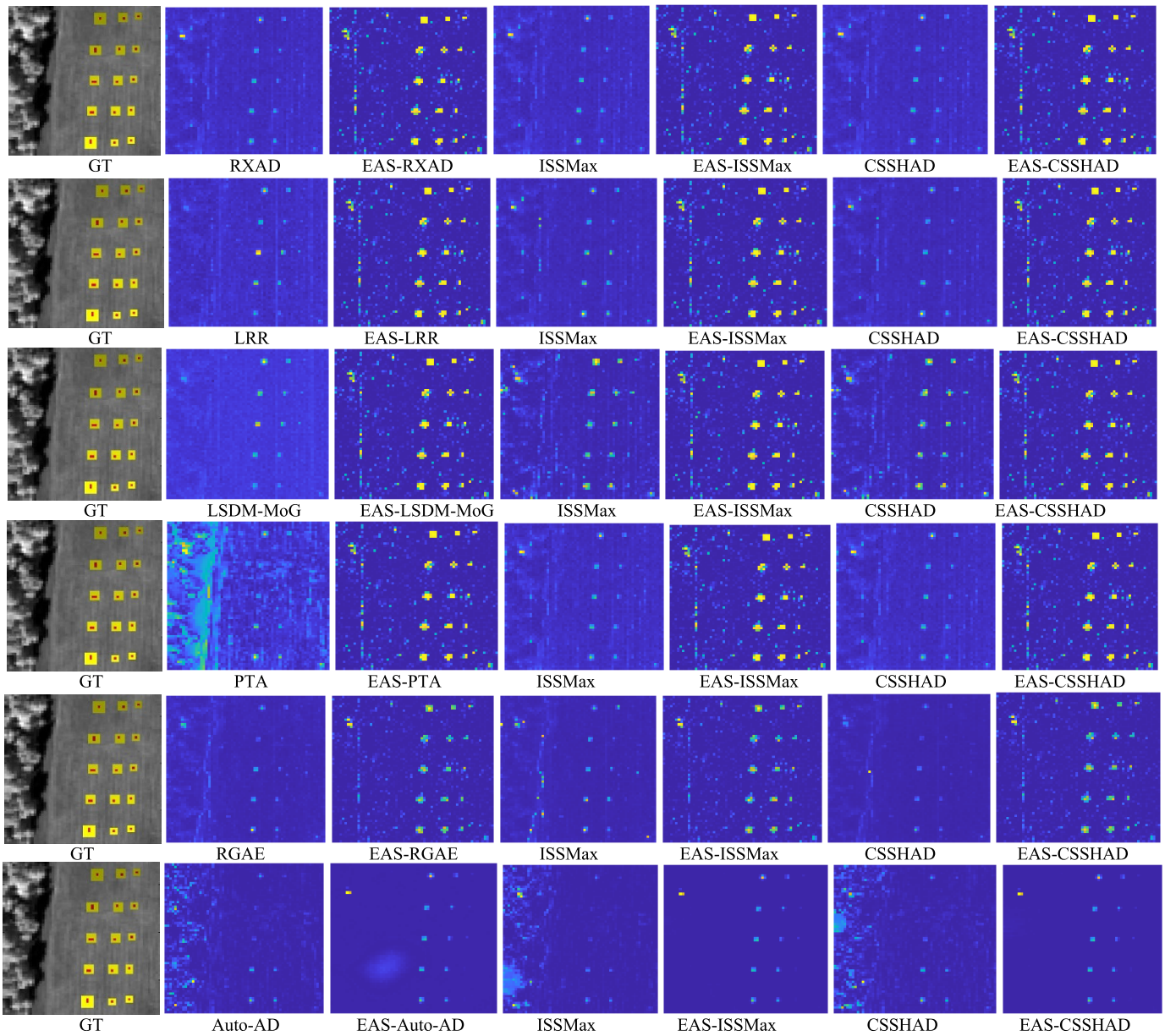


Fig. 6. GT maps and ADMaps produced by RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD, along with their corresponding EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts, using RXAD as the anomaly detector for the HYDICE panel scene.

TABLE III

EIGHT DETECTION MEASURES IN ADDRESSING FOUR ISSUES, BS, ANOMALY DETECTABILITY, NOISE EFFECT, INTERBAND CORRELATION REDUCTION

Detection measure	BKG suppression	Anomaly detectability	Noise effect	Inter-band correlation reduction
$AUC_{(D,F)}$	√	√		
AUC_{ADP}		√		
AUC_{BDP}			√	
AUC_{JAD}		√		√
AUC_{JBS}	√			√
AUC_{ADBS}	√	√		
AUC_{OADP}	√	√	√	
AUC_{SBPR}	√	√	√	√

Fig. 4(a) shows the pseudo-color image. The original image is 400 × 400 pixels, with a spatial resolution of 3.5 m, and includes 224 spectral channels that cover wavelengths from 370 to 2510 nm. After excluding bands related to water absorption, those with low signal-to-noise ratios (SNRs), and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 bands were available for the experiments. An area of size 100 × 100 pixels was cropped from the upper left corner of the scene as the test image shown in Fig. 4(b), with its GT map given in Fig. 4(c) obtained by visual inspection.

The selected scene is commonly used to evaluate the performance of HAD. As shown in Fig. 4(b), the proclaimed anomalies are the three airplanes in the upper left corner, which cover 85 pixels and approximately 0.33% of the whole image. However, this scene is more suitable for target detection than anomaly detection due to the three aircrafts of large sizes in pixels on the scene. As a result, the conclusions drawn from this scene differ significantly from those in Figs. 2 and 3.

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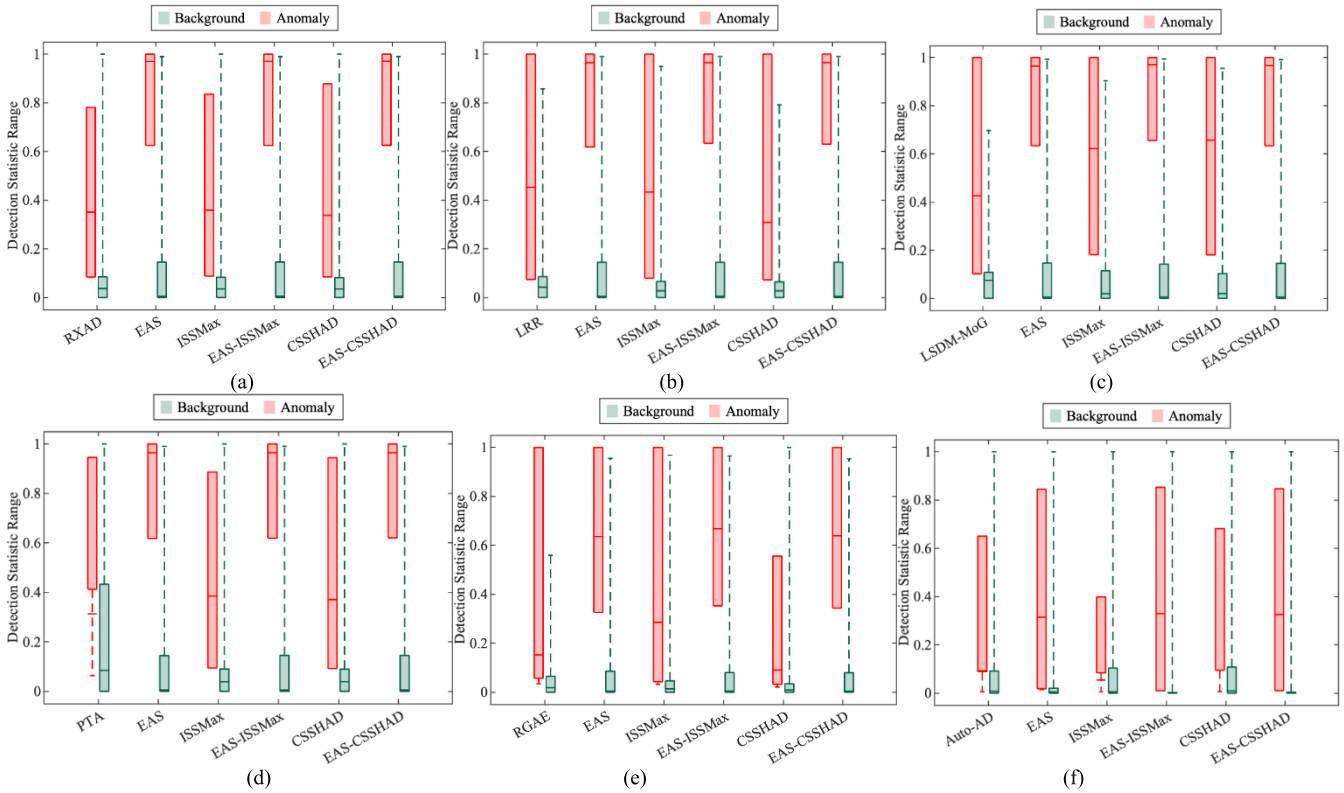


Fig. 7. BKG-anomaly SMs of (a) RXAD, (b) LRR, (c) LSDM-MoG, (d) PTA, (e) RGAE, (f) Auto-AD, and their corresponding EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts using RXAD as the anomaly detector for the HYDICE panel scene.

D. AVIRIS Bay Champagne Beach Scene

A fourth dataset acquired by the AVIRIS sensor is the Bay Champagne beach scene. It was captured on July 4, 2010, and has a size of 100×100 pixels. After removing the noisy bands from the original dataset, 188 bands remained. The spatial resolution is approximately 4.4 m. The top-central pixels are considered the primary anomalous pixels. Fig. 5 shows the pseudo-color map and GT map of the Bay Champagne beach scene.

V. Experimental Results and Discussions

Although CSSHAD can be implemented with any arbitrary anomaly detector, it is impossible to implement all available anomaly detectors. So, we have particularly selected six recently developed anomaly detectors, LRR [31], LSDM-MoG [47], PTA [49], RGAE [17], and Auto-AD [16], along with classical RXAD [19] for comprehensive study. The reason that the selection of these methods can be also found in [3] and was based on the fact that each of them represents different categories. LRR uses a low-rank approximation to separate anomalies from BKG, while LSDM-MoG employs Gaussian distributions for BKG estimation. PTA utilizes tensor decomposition. RGAE is a graph-based method assuming a Gaussian distribution for BKG. Auto-AD is a DL method which is based on AE. These methods used different distance measures, such as Mahlabonis distance, Euclidean distance, and Manhattan distance, we unified all these distances to use Mahalanobis distance for consistency because RXAD indeed implements as Mahalanobis distance.

A comparative analysis was conducted via extensive experiments on implementing these seven anomaly detectors in CSSHAD and EAS-CSSHAD on the hyperspectral data scenes described in Section IV.

Table II lists the parameter settings for the AD methods used for the experiments which were performed under MATLAB R2024b with macOS Sequoia version 15.3, CPU chip Apple M4, ten cores, and 24-GB memory.

The experiments were performed under MATLAB R2024b with macOS Sequoia version 15.3, CPU chip Apple M4, ten cores, and 24-GB memory.

A. Quantitative Detection Measures

In HAD, the absence of prior target knowledge significantly hampers HAD performance due to BKG, noise, and interference. The conventional 2-D ROC analysis can only evaluate detector effectiveness but fails to address AD, BS. Hence, the concept of 3-D ROC curves was first introduced in [54], which include a third parameter τ in addition to P_D and P_F to construct a 3-D (P_D, P_F, τ) curve from three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) can be plotted and their areas under these 2-D ROC curves (AUC) can be further calculated as $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$ and $\text{AUC}_{(F,\tau)}$, that can be used to derive eight detection measures [53], [54], [55] as follows.

1) Effectiveness of a Detector: The area under a 2-D ROC curve of (P_D, P_F), $\text{AUC}_{(D,F)}$ with

$$0 \leq \text{AUC}_{(D,F)} \leq 1 \tag{7}$$

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TABLE IV

AUC VALUES PRODUCED BY RXAD, LRR, LSDM-MOG, PTA, RGAE, AND AUTO-AD ALONG WITH THEIR CORRESPONDING EAS, ISSMAX, EAS-ISSMAX, CSSHAD, AND EAS-CSSHAD COUNTERPARTS USING RXAD AS THE ANOMALY DETECTOR FOR THE HYDICE PANEL SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SBPR}	AUC_{OADP}	Time(s)
RXAD	--	--	0.99	0.3574	0.9573	1.3474	1.9473	1.3147	8.3663	2.3047	0.031
EAS-RXAD	--	--	0.9932	0.9227	0.9627	1.9159	1.9559	1.8854	24.7254	2.8786	0.023
ISSMax-RXAD	1	0.976	0.9918	0.3618	0.9584	1.3537	1.9502	1.3202	8.6891	2.312	1.346
EAS-ISSMax-RXAD	1	0.992	0.9932	0.9233	0.9625	1.9165	1.9557	1.8858	24.5959	2.879	1.322
CSSHAD-RXAD	1	0.953	0.9918	0.3581	0.9589	1.3499	1.9507	1.317	8.7065	2.3088	2.163
EAS-CSSHAD-RXAD	1	0.977	0.9932	0.9239	0.9624	1.9171	1.9557	1.8863	24.5986	2.8795	2.333
LRR	--	--	0.9911	0.4296	0.9519	1.4207	1.943	1.3815	8.9338	2.3726	14.849
EAS-LRR	--	--	0.9933	0.9183	0.9628	1.9116	1.9561	1.8811	24.7028	2.8744	1.947
ISSMax-LRR	5	0.978	0.9928	0.4064	0.9661	1.3992	1.9589	1.3725	11.9857	2.3653	82.441
EAS-ISSMax-LRR	1	0.969	0.9934	0.9202	0.9627	1.9135	1.9561	1.8829	24.6697	2.8763	7.41
CSSHAD-LRR	20	0.95	0.9916	0.2983	0.9672	1.2899	1.9588	1.2655	9.0832	2.2571	297.345
EAS-CSSHAD-LRR	1	0.969	0.9932	0.9203	0.9627	1.9135	1.9559	1.883	24.6502	2.8762	7.93
LSDM-MoG	--	--	0.9928	0.4351	0.9213	1.4279	1.9141	1.3564	5.5304	2.3492	5.731
EAS-LSDM-MoG	--	--	0.9932	0.928	0.9627	1.9213	1.9559	1.8907	24.8801	2.8839	8.946
ISSMax-LSDM-MoG	6	0.965	0.9926	0.5909	0.9618	1.5835	1.9544	1.5527	15.4721	2.5453	43.397
EAS-ISSMax-LSDM-MoG	2	0.977	0.993	0.9331	0.9623	1.926	1.9552	1.8954	24.7323	2.8884	30.065
CSSHAD-LSDM-MoG	8	0.951	0.9927	0.5847	0.9629	1.5774	1.9556	1.5476	15.7723	2.5403	70.758
EAS-CSSHAD-LSDM-MoG	1	0.976	0.9927	0.9214	0.9624	1.9142	1.9552	1.8838	24.5242	2.8765	19.522
PTA	--	--	0.8315	0.3894	0.8632	1.221	1.6948	1.2526	2.8476	2.0841	27.81
EAS-PTA	--	--	0.9933	0.9177	0.9627	1.911	1.956	1.8804	24.6078	2.8737	25.969
ISSMax-PTA	1	1	0.9915	0.3862	0.9548	1.3778	1.9464	1.341	8.5529	2.3325	53.336
EAS-ISSMax-PTA	1	0.969	0.9933	0.9182	0.9625	1.9115	1.9558	1.8807	24.4739	2.874	53.367
CSSHAD-PTA	1	0.955	0.9918	0.3822	0.9549	1.374	1.9466	1.3371	8.466	2.3289	55.818
EAS-CSSHAD-PTA	1	1	0.9933	0.919	0.9625	1.9123	1.9558	1.8815	24.4851	2.8748	48.984
RGAE	--	--	0.9643	0.2078	0.9748	1.1721	1.939	1.1826	8.2304	2.1469	31.758
EAS-RGAE	--	--	0.9945	0.654	0.9764	1.6486	1.971	1.6304	27.7701	2.6249	25.68
ISSMax-RGAE	20	0.867	0.9771	0.2672	0.9787	1.2443	1.9559	1.2459	12.5691	2.223	583.63
EAS-ISSMax-RGAE	1	0.966	0.9944	0.6619	0.9765	1.6563	1.9709	1.6384	28.1834	2.6328	52.702
CSSHAD-RGAE	20	0.167	0.9705	0.1572	0.9869	1.1277	1.9574	1.1441	12.0171	2.1146	533.127
EAS-CSSHAD-RGAE	1	0.967	0.9941	0.646	0.9767	1.6401	1.9708	1.6227	27.7466	2.6168	46.445
Auto-AD	--	--	0.8643	0.181	0.9774	1.0454	1.8418	1.1584	8.016	2.0227	17.086
EAS-Auto-AD	--	--	0.9886	0.3227	0.992	1.3112	1.9806	1.3147	40.3768	2.3033	4.761
ISSMax-Auto-AD	20	0.219	0.8587	0.1088	0.979	0.9675	1.8377	1.0878	5.1893	1.9465	304.947
EAS-ISSMax-Auto-AD	1	1	0.9976	0.3232	0.9983	1.3208	1.9959	1.3215	188.2405	2.3191	11.638
CSSHAD-Auto-AD	20	0.251	0.8681	0.1988	0.9742	1.0668	1.8422	1.173	7.6931	2.0411	286.959
EAS-CSSHAD-Auto-AD	1	1	0.9972	0.3212	0.9979	1.3184	1.9951	1.3191	153.6708	2.3163	9.453

can be used to evaluate detector efficacy but cannot evaluate AD and BS.

2) Anomaly Detection Probability: The area under a 2-D ROC curve of (P_D, τ), is calculated AUC_{ADP} for ADP with

$$0 \leq AUC_{ADP} \leq 1. \tag{8}$$

3) Joint Anomaly Detectability (JAD): Combining $AUC_{(D,F)}$ in (7) and AUC_{ADP} in (8) yields the JAD, AUC_{JAD}

$$0 \leq AUC_{JAD} = AUC_{(D,F)} + AUC_{ADP} \leq 2. \tag{9}$$

4) BKG Detection Probability: While P_F represents false alarm probability for TD, $1–P_F$ represents BDP, AUC_{BDP} given by

$$0 \leq AUC_{BDP} = 1 - AUC_{(F,\tau)} \leq 1. \tag{10}$$

Like JAD, combining AUC_{BDP} in (10) with $AUC_{(D,F)}$ in (7) also yields joint BS (JBS), AUC_{JBS}

$$0 \leq AUC_{JBS} = AUC_{(D,F)} + AUC_{BDP} \leq 2. \tag{11}$$

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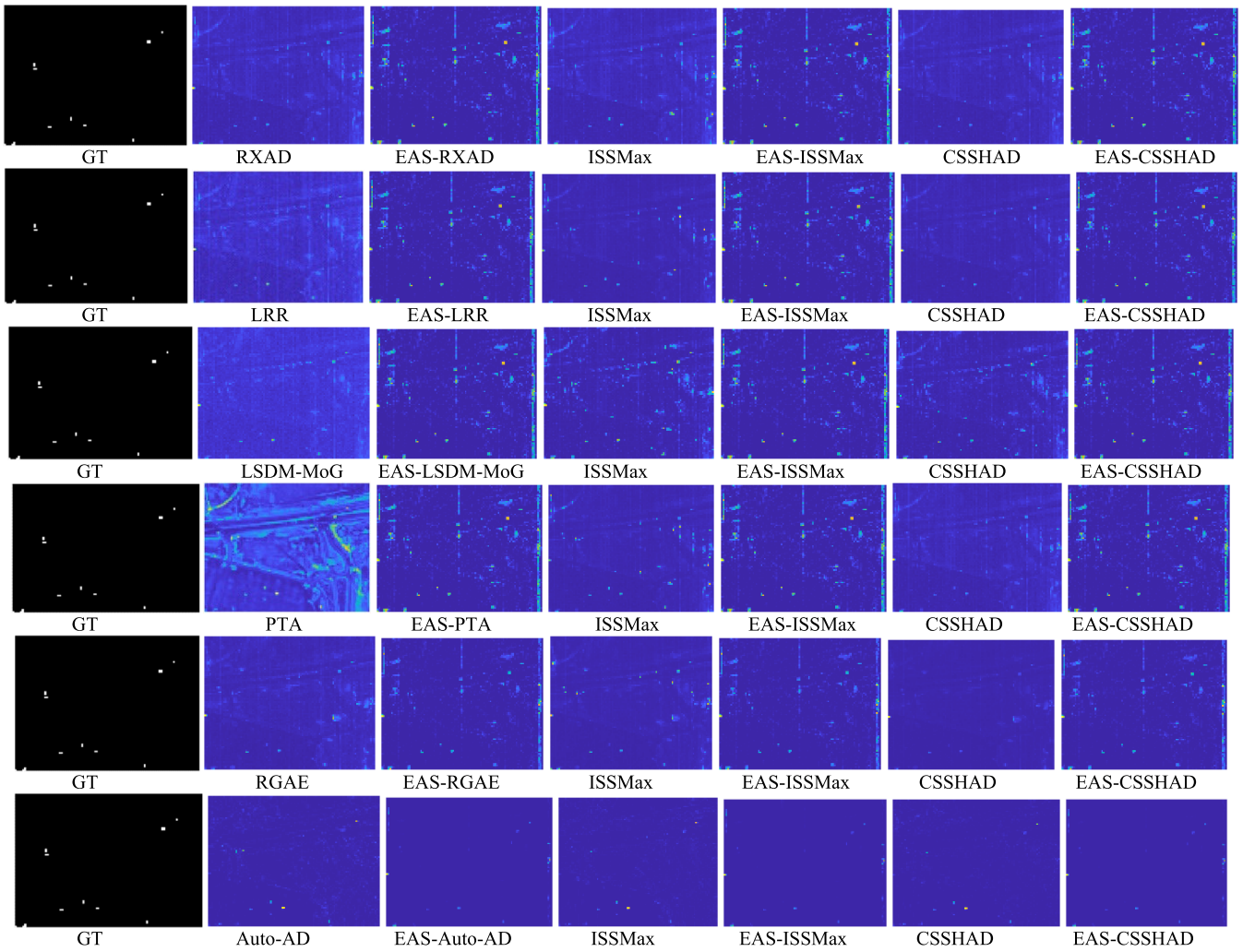


Fig. 8. GT maps and ADMaps produced by RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD, along with their corresponding EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts, using RXAD as the anomaly detector for the HYDICE urban scene.

5) JAD With BS: $\text{AUC}_{(D,F)}$ measures, P_D, and P_F lack the nuances of ADP and BDP. Factoring $\text{AUC}_{(F,\tau)}$ into $\text{AUC}_{(D,\tau)}$ requires innovative metrics as follows.

a) AD in BKG (ADBS): To measure both AD and BS, we define AUC_{ADBS} by emphasizing anomaly detection and BS as follows:

$$0 \leq \text{AUC}_{\text{ADBS}} = \text{AUC}_{\text{ADP}} + \text{AUC}_{\text{BDP}} = \text{AUC}_{\text{OA}} \leq 2. \quad (12)$$

b) Overall anomaly detection power: Now combining AUC_{ADBS} in (12) with $\text{AUC}_{(D,F)}$ in (7) results in AUC_{OADP} given by

$$0 \leq \text{AUC}_{\text{OADP}} = \text{AUC}_{(D,F)} + \text{AUC}_{\text{ADP}} + \text{AUC}_{\text{BDP}} \leq 3. \quad (13)$$

c) Signal-to-BKG probability ratio (SBPR): The SBPR similar to the SNR can be defined by the ratio of ADP to BDP, AUC_{BDP} given by

$$0 \leq \text{AUC}_{\text{SBPR}} = \frac{\text{AUC}_{\text{ADP}}}{\text{AUC}_{\text{BDP}}}. \quad (14)$$

Table III presents eight detection measures addressing the four significant AD issues highlighted in the abstract.

In Table III, a check symbol ($\surd$) indicates the effectiveness of a measure in addressing a specific issue. A red check symbol ($\surd$ in red) signifies a strong "yes," indicating that a measure can address a particular issue effectively, whereas a black check symbol ($\surd$ in black) indicates a weak "yes," suggesting that a measure is less effective in address a particular issue but still has a positive response to the corresponding issue.

B. HYDICE 15-Panel Scene

Fig. 6 shows the GT maps and ADMaps of RXAD, LRR, LSDM-MoG, PTA, RGAE, Auto-AD, along with their corresponding counterparts, EAS, ISSMax which is a particular version of ISSHAD developed in [11], EAS-ISSMax, CSSHAD, and EAS-CSSHAD for the HYDICE panel scene.

Two findings are very evident by visual assessment. One is that both ISSMax and CSSHAD performed very comparably and generally performed better than or at least as well as original anomaly detectors, specifically, LSDM-MoG and PTA. The other is that EAS significantly improved anomaly detectability of all anomaly detectors by detecting all 15 panels as anomalies, while suppressing BKG effectively.

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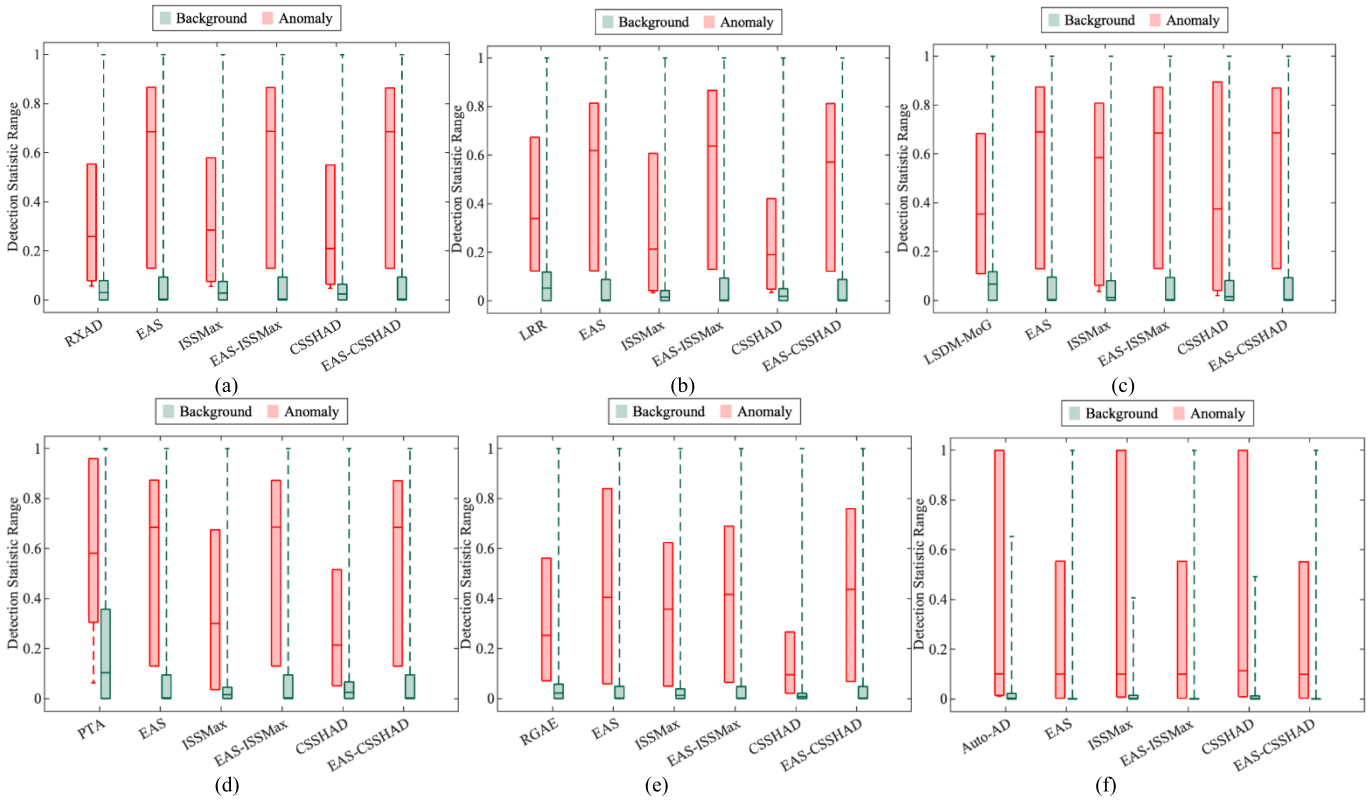


Fig. 9. BKG-anomaly SMs of (a) RXAD, (b) LRR, (c) LSDM-MoG, (d) PTA, (e) RGAE, and (f) Auto-AD, and their corresponding EAS, ISSMax, CSSHAD, and EAS-CSSHAD counterparts using RXAD as the anomaly detector for the HYDICE urban scene.

The visual inspection of the detection maps (see Fig. 6) did not tell how much degree of anomalies separated from BKG. To see anomaly-BKG separability (ABS), the separability map (SM) developed in [56] was used to plot two boxes, the anomaly box and the BKG box. Fig. 7(a)–(f) shows SMs of RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD along with their counterparts operating in EAS, ISSMax, EAS-ISSMax, CSSHAD, EAS-CSSHAD, where represent anomaly and BKG boxes highlighted by red and green, respectively, were obtained by specifying the range of [10%, 90%], such as [57] corresponding to a particular threshold value $\tau = 10\%$. According to Fig. 7, it clearly shows that all the six anomaly detectors implemented in EAS significantly improved the separation of anomalies from BKG in terms of ABS. Overall speaking, EAS-CSSHAD seemed to perform the best.

All the conclusions were based on visual examination from Figs. 6 and 7. Apparently, such qualitative analysis may not be reliable when it comes to the case that two anomaly detectors which performed very closely, such as ISSMax and CSSHAD, with/without operating in EAS. In this case, a quantitative analysis is necessary. So, the AUC-derived detection measures developed in Section V-A can be used for this purpose. To do so, the 3-D ROC curves of (P_D, P_F, τ) along with three 2-D ROC curves of (P_D, P_F), (P_D, τ) and (P_F, τ) were generated from Fig. 6. Since what we are interested in is quantitative analysis using the eight anomaly detection measures, the ROC curves are not included in this article to save space. Instead, Table IV calculates the eight AUC values, $\text{AUC}_{(D,F)}$, AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SBPR}, and AUC_{OADP} defined in (7)–(14) for quantitative comparison, where the best results are highlighted by red. It should be noted that if two anomaly detectors produced the best results with very close AUC values within negligible differences, they were highlighted by red as well. Also documented in Table IV are the computing times and the number of iterations, j, along with $\text{TI}_{j,j-1}$ values.

According to Table IV, CSSHAD has improved all the anomaly detectors RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD, as evidenced by higher $\text{AUC}_{(D,F)}$ scores. By operating all the compared six anomaly detectors in EAS, their anomaly detection performances can be further enhanced. In particular, EAS-CSSHAD-RXAD, EAS-CSSHAD-LRR, and EAS-CSSHAD-PTA. EAS-CSSHAD significantly outperformed CSSHAD in detecting anomalies while maintaining effective BKG suppression by improving AUC_{SNPR} values. In addition, EAS-CSSHAD methods balanced performance and computational efficiency more effectively. Operating CSSHAD in EAS yielded superior performance to ISSMax and CSSHAD and also worked at least as well as or better than EAS-ISSMax.

C. HYDICE Urban Scene

Fig. 8 shows GT maps and ADMaps of RXAD, LRR, LSDM-MoG, PTA, RGAE, Auto-AD, along with their corresponding counterparts, EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD for the HYDICE urban scene. By visual inspection of Fig. 8, it is very difficult to assess the detection performance because BKG is very complicated and affects anomaly detection. This can be seen from the

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TABLE V

AUC VALUES PRODUCED BY RXAD, LRR, LSDM-MOG, PTA, RGAE, AND AUTO-AD ALONG WITH THEIR CORRESPONDING EAS, ISSMAX, EAS-ISSMAX, CSSHAD, AND EAS-CSSHAD COUNTERPARTS USING RXAD AS THE ANOMALY DETECTOR FOR THE HYDICE URBAN SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SBPR}	AUC_{OADP}	Time(s)
RXAD	--	--	0.9874	0.2575	0.9641	1.2449	1.9514	1.2216	7.1647	2.209	0.046
EAS-RXAD	--	--	0.9957	0.6439	0.9767	1.6395	1.9724	1.6206	27.6419	2.6163	0.04
ISSMax-RXAD	5	0.965	0.9865	0.2758	0.9647	1.2623	1.9511	1.2405	7.8076	2.227	10.355
EAS-ISSMax-RXAD	1	0.996	0.9956	0.6442	0.9766	1.6398	1.9722	1.6208	27.5568	2.6164	3.469
CSSHAD-RXAD	1	0.953	0.9918	0.3581	0.9589	1.3499	1.9507	1.317	8.7065	2.3088	2.163
EAS-CSSHAD-RXAD	1	0.996	0.9956	0.6441	0.9767	1.6397	1.9723	1.6208	27.5875	2.6164	4.156
LRR	--	--	0.9921	0.3692	0.9413	1.3613	1.9335	1.3105	6.2941	2.3026	23.495
EAS-LRR	--	--	0.9955	0.6064	0.9784	1.6019	1.9739	1.5848	28.0225	2.5803	9.166
ISSMax-LRR	11	0.951	0.9884	0.2352	0.98	1.2237	1.9685	1.2152	11.7854	2.2036	380.183
EAS-ISSMax-LRR	1	0.958	0.9954	0.6341	0.9769	1.6295	1.9723	1.611	27.4985	2.6064	22.65
CSSHAD-LRR	20	0.95	0.9916	0.2983	0.9672	1.2899	1.9588	1.2655	9.0832	2.2571	297.345
EAS-CSSHAD-LRR	1	0.996	0.9954	0.6025	0.9784	1.5979	1.9739	1.5809	27.9404	2.5763	25.031
LSDM-MoG	--	--	0.9915	0.3741	0.9281	1.3656	1.9196	1.3022	5.2053	2.2937	8.59
EAS-LSDM-MoG	--	--	0.9956	0.6459	0.9765	1.6416	1.9722	1.6224	27.5301	2.618	9.166
ISSMax-LSDM-MoG	20	0.772	0.9867	0.4821	0.9754	1.4688	1.9621	1.4575	19.5869	2.4442	268.721
EAS-ISSMax-LSDM-MoG	1	0.979	0.9956	0.6468	0.9764	1.6424	1.9721	1.6232	27.4591	2.6188	31.56
CSSHAD-LSDM-MoG	8	0.951	0.9927	0.5847	0.9629	1.5774	1.9556	1.5476	15.7723	2.5403	70.758
EAS-CSSHAD-LSDM-MoG	1	0.975	0.9956	0.6463	0.9765	1.6419	1.9721	1.6228	27.4846	2.6184	33.484
PTA	--	--	0.916	0.5504	0.8617	1.4664	1.7777	1.4121	3.9807	2.3281	46.656
EAS-PTA	--	--	0.9956	0.6453	0.9767	1.6409	1.9724	1.622	27.7259	2.6176	48.139
ISSMax-PTA	20	0.978	0.9887	0.2959	0.9777	1.2846	1.9664	1.2736	13.24	2.2623	1191.367
EAS-ISSMax-PTA	1	1	0.9956	0.6456	0.9766	1.6412	1.9722	1.6222	27.6138	2.6178	103.459
CSSHAD-PTA	1	0.955	0.9918	0.3822	0.9549	1.374	1.9466	1.3371	8.466	2.3289	55.818
EAS-CSSHAD-PTA	1	0.991	0.9956	0.6461	0.9767	1.6417	1.9723	1.6228	27.6794	2.6184	102.351
RGAE	--	--	0.9946	0.2519	0.9738	1.2464	1.9683	1.2257	9.5966	2.2203	47.134
EAS-RGAE	--	--	0.9958	0.4263	0.9869	1.4221	1.9826	1.4132	32.48	2.409	51.018
ISSMax-RGAE	20	0.9	0.9939	0.3471	0.9827	1.341	1.9765	1.3298	20.0142	2.3237	1096.914
EAS-ISSMax-RGAE	4	0.964	0.9955	0.3989	0.987	1.3944	1.9825	1.3859	30.7671	2.3814	284.532
CSSHAD-RGAE	20	0.167	0.9705	0.1572	0.9869	1.1277	1.9574	1.1441	12.0171	2.1146	533.127
EAS-CSSHAD-RGAE	20	0.917	0.9963	0.4357	0.987	1.432	1.9833	1.4227	33.43	2.419	959.092
Auto-AD	--	--	0.981	0.2349	0.9934	1.2159	1.9744	1.2283	35.3186	2.2093	56.865
EAS-Auto-AD	--	--	0.9955	0.1459	0.9988	1.1414	1.9943	1.1447	121.4821	2.1402	9.29
ISSMax-Auto-AD	7	1	0.9786	0.2195	0.9955	1.1981	1.974	1.215	48.5075	2.1936	525.155
EAS-ISSMax-Auto-AD	5	1	0.9953	0.1462	0.9988	1.1416	1.9941	1.145	119.8713	2.1403	58.623
CSSHAD-Auto-AD	8	1	0.9877	0.2098	0.9962	1.1975	1.9839	1.206	55.4078	2.1937	575.197
EAS-CSSHAD-Auto-AD	2	1	0.9956	0.1456	0.9988	1.1412	1.9944	1.1444	121.7458	2.14	27.799

detection results of all anomaly detectors operating in EAS which suppressed BKG reasonably well, such as EAS-RXAD and EAS-CSSHAD.

Fig. 9(a)–(f) shows SMs of RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD along with their counterparts operating in EAS, ISSMax, EAS-ISSMax, CSSHAD, EAS-CSSHAD, where represent anomaly and BKG boxes highlighted by red and green, respectively, were obtained by specifying the range of [10%, 90%] corresponding to a particular percentile value $\tau = 10\%$. Fig. 9 also confirmed the difficulty with the observations made from Fig. 8. However, Fig. 9 did clearly show that an anomaly operating in EAS could actually completely separated anomalies from BKG as shown by the nonoverlapped gaps between anomaly and BKG boxes. The best performance for ABS is clearly EAS-CSSHAD.

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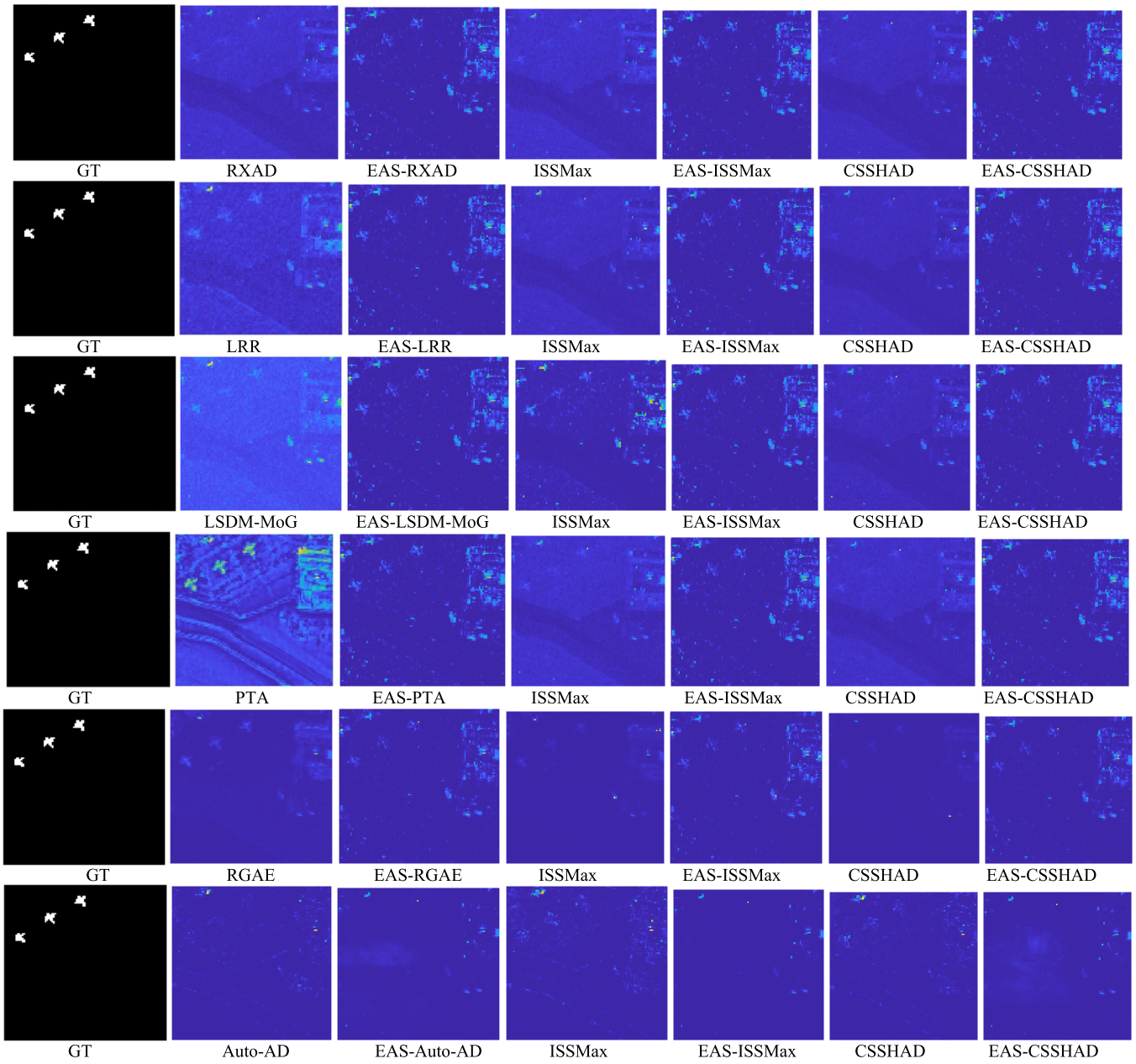


Fig. 10. GT maps and ADMaps produced by RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD, along with their corresponding EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts, using RXAD as the anomaly detector for the AVIRIS San Diego airport scene.

EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts using RXAD as the anomaly detector for the HYDICE urban scene.

Apparently, using Figs. 8 and 9 for qualitative analysis are not reliable. Particularly, when two anomaly detectors produced the best results with very close AUC values within negligible differences. So, in this case, the 3-D ROC curves of (P_D, P_F, τ) along with three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) were generated to calculate the eight anomaly detection measures defined in (7)–(14). To save space, the ROC curves are not included here. Instead, Table V calculates the eight ROC curve-derived AUC values, $\text{AUC}_{(D,F)}$, AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SBPR}, and AUC_{OADP} for quantitative comparison with the best results highlighted by red, where if two anomaly detectors produced best results with very close AUC values within negligible differences, they were highlighted by red as well. Also documented in Table V are the computing times and the number of iterations, j, along with $\text{TI}_{j,j-1}$ values.

According to Table V, EAS-based anomaly detectors improved their counterparts without using EAS in all the eight detection measures. Except for LRR and Auto-AD, EAS-CSSHAD was the best.

D. AVIRIS San Diego Airport Scene

As noted in Fig. 4, the three airplanes in the scene are highly visible and thus hardly considered anomalies due to their large sizes. The issue in this data scene is the GT which is based on visual inspection via human naked eyes. So, the true anomalies

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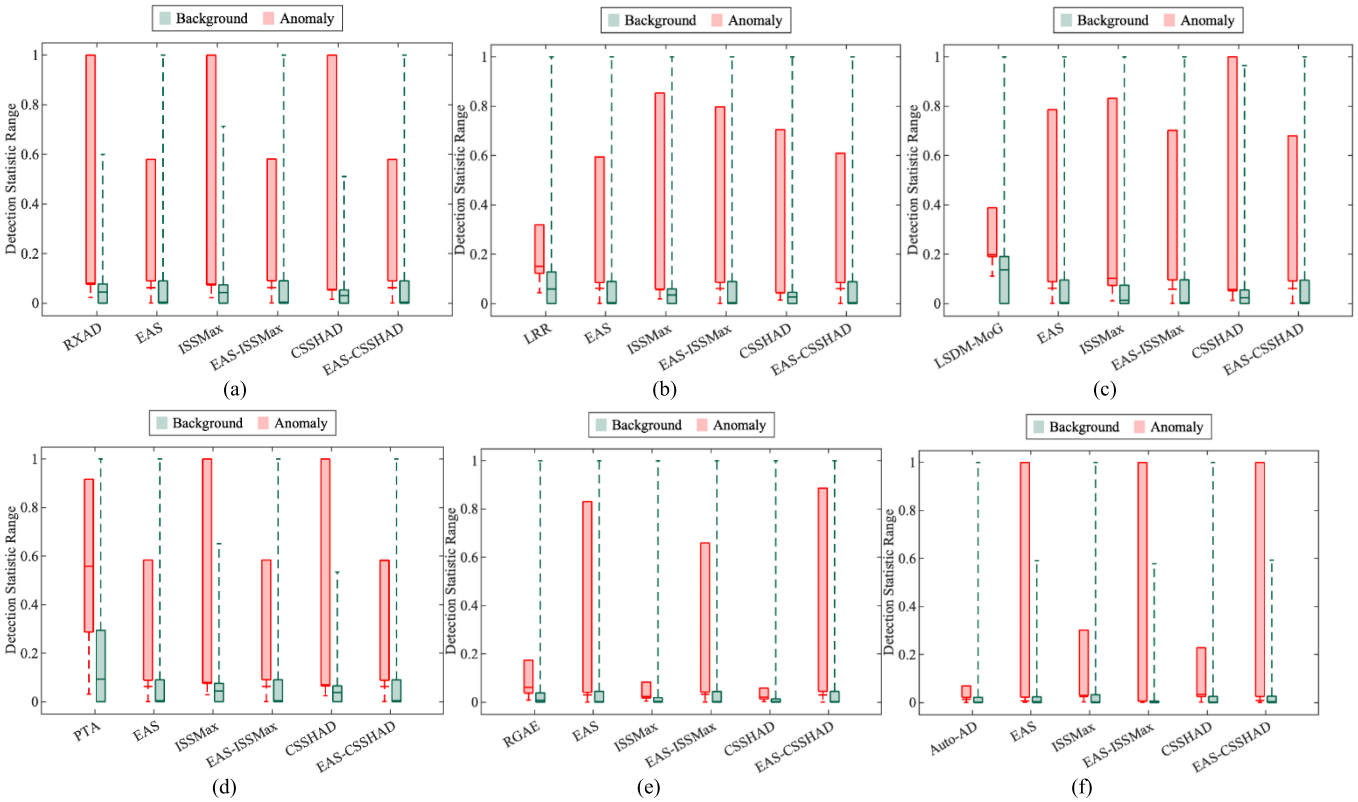


Fig. 11. BKG-anomaly SMs of (a) RXAD, (b) LRR, (c) LSDM-MoG, (d) PTA, (e) RGAE, and (f) Auto-AD, and their corresponding EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts using RXAD as the anomaly detector for the AVIRIS San Diego airport scene.

may actually occur at the right upper corner not these three airplanes. Consequently, this scene is more suitable for TD than anomaly detection, and the conclusions derived from this scene are quite different from those in previous experiments.

Fig. 10 shows GT maps and ADMaps of RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD, along with their corresponding counterparts, EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD for this scene. As can be seen from Fig. 10, RXAD barely detected the three airplanes but rather detected unknown targets at the upper right corner. So did CSSHAD. By contrast, LRR and PTA clearly detected the three airplanes. So, if RXAD and CSSHAD are anomaly detectors, then LRR and PTA cannot be anomaly detectors. Conversely, if LRR and PTA are anomaly detectors, then RXAD and CSSHAD cannot be anomaly detectors. This led to a confusing conclusion that which detector was a real anomaly detector. These results showed that the provided GT was not reliable. Operating an anomaly detector in EAS simply manifested the effects.

To conduct a further study. Fig. 11(a)–(f) shows SMs of RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD along with their counterparts operating in EAS, ISSMax, EAS-ISSMax, CSSHAD, EAS-CSSHAD, where represent anomaly and BKG boxes highlighted by red and green, respectively, and were obtained by specifying the range of [10%, 90%] corresponding to a particular threshold value $\tau = 10\%$. According to Fig. 11(a)–(f), the anomaly box and BKG box were barely separated by visual inspection. But how close the gaps were could not be determined quantitatively by visual inspection of Fig. 11(a)–(f).

Like Tables III and IV, Table VI calculates the eight AUC values, $\text{AUC}_{(D,F)}$, AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SBPR}, and AUC_{OADP} for quantitative comparison, where if two anomaly detectors produced the best results with very close AUC values within negligible differences, they were highlighted in red as well. Also documented in Table VI are the computing times and the number of iterations, j, along with $\text{TI}_{j,j-1}$ values. From Table VI, LRR and PTA without operating in EAS produced the best results which pretty much confirmed the detection results in Fig. 12 based on the given GT which assumed that three airplanes were anomalies. These contradicted experiments demonstrated that selecting an appropriate dataset for HAD is crucial. This issue was discussed in [4] and extensively in [3].

E. AVIRIS Bay Champagne Beach Scene

Unlike the San Diego airport scene which has three large airplanes, the anomaly in this scene pretty much stands alone by itself in the scene. Fig. 12 shows GT maps and ADMaps of RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD, along with their corresponding counterparts, EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD for the Bay Champagne beach scene. However, there is an issue associated with this scene. It is clear that there are also some anomalies on the beach which cannot be identified and are included in the GT map, as shown in Fig. 12, where some anomalies on the beach were also detected. This is indeed the case where the detection results are produced by anomaly detectors operating in EAS.

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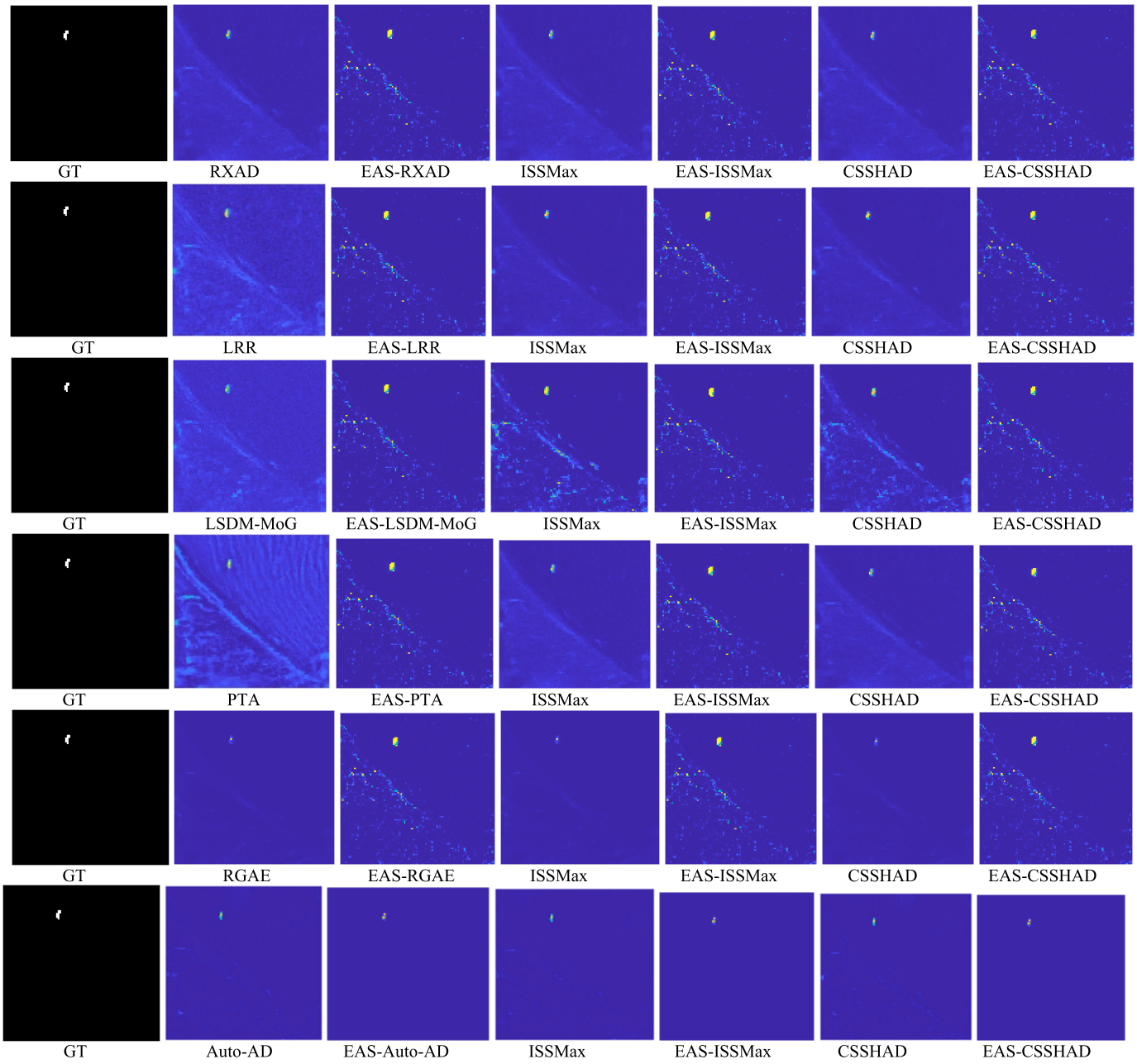


Fig. 12. GT maps and ADMaps produced by RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD, along with their corresponding EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts, using RXAD as the anomaly detector for the AVIRIS Bay Champagne beach scene.

Fig. 13(a)–(f) shows SMs of RXAD, LRR, LSDM-MoG, PTA, RGAE, and Auto-AD along with their counterparts operating in EAS, ISSMax, EAS-ISSMax, CSSHAD, EAS-CSSHAD, where represent anomaly and BKG boxes highlighted by red and green, respectively, were obtained by specifying the range of [10%, 90%] corresponding to a particular threshold value $\tau = 10\%$. It clearly showed from Fig. 13 that the anomaly box and BKG box were completely separated, and EAS could further enhance ABS by enlarging the gaps between the two boxes. Also, by visually inspecting the largest gaps between two boxes, EAS-CSSHAD was the best.

To conduct a quantitative study, 3-D ROC curves of (P_D, P_F, τ) along with three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ) were generated from Fig. 12 to calculate the eight anomaly detection measures defined in (7)–(14). To save space, the ROC curves are not included in this article. Instead, Table VII calculates the eight AUC values, $\text{AUC}_{(D,F)}$, AUC_{ADP}, AUC_{BDP}, AUC_{JAD}, AUC_{JBS}, AUC_{ADBS}, AUC_{SBPR}, and AUC_{OADP} for quantitative comparison, where if two anomaly detectors produced the best results with very close AUC values within negligible differences, they were highlighted in red as well. Also documented in Table VII are the computing times and the number of iterations, j, along with $\text{TI}_{j,j-1}$ values.

According to Table VII, the best results were always the ones implementing EAS with different versions. For example, RXAD, LRR, LSDM-MoG, PTA, RGAE, and

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TABLE VI

AUC VALUES PRODUCED BY RXAD, LRR, LSDM-MOG, PTA, RGAE, AND AUTO-AD ALONG WITH THEIR CORRESPONDING EAS, ISSMAX, EAS-ISSMAX, CSSHAD, AND EAS-CSSHAD COUNTERPARTS USING RXAD AS THE ANOMALY DETECTOR FOR THE AVIRIS SAN DIEGO AIRPORT SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SBPR}	AUC_{OADP}	Time(s)
RXAD	--	--	0.8314	0.0802	0.9547	0.9116	1.7861	1.0349	1.7711	1.8663	0.064
EAS-RXAD	--	--	0.8512	0.0966	0.98	0.9477	1.8312	1.0766	4.8375	1.9278	0.063
ISSMax-RXAD	1	0.964	0.8317	0.0765	0.9566	0.9082	1.7883	1.0331	1.7612	1.8648	4.942
EAS-ISSMax-RXAD	1	0.983	0.8512	0.0972	0.98	0.9484	1.8311	1.0772	4.8496	1.9284	5.006
CSSHAD-RXAD	1	1	0.8314	0.0554	0.9686	0.8868	1.8	1.024	1.7665	1.8554	5.53
EAS-CSSHAD-RXAD	1	0.997	0.8512	0.097	0.9799	0.9482	1.8311	1.0769	4.8364	1.9281	5.722
LRR	--	--	0.9235	0.1558	0.9324	1.0793	1.8559	1.0882	2.3049	2.0117	27.684
EAS-LRR	--	--	0.8511	0.0938	0.9806	0.9449	1.8317	1.0744	4.8275	1.9255	17.618
ISSMax-LRR	1	0.964	0.7953	0.0681	0.9641	0.8635	1.7595	1.0322	1.8999	1.8275	68.69
EAS-ISSMax-LRR	1	0.974	0.8513	0.0966	0.9807	0.9478	1.832	1.0773	5.0079	1.9286	40.468
CSSHAD-LRR	3	1	0.8093	0.0559	0.9726	0.8652	1.7819	1.0285	2.0371	1.8378	124.34
EAS-CSSHAD-LRR	1	0.959	0.8514	0.0939	0.9807	0.9453	1.832	1.0746	4.8531	1.926	35.231
LSDM-MoG	--	--	0.8403	0.2111	0.8571	1.0513	1.6973	1.0682	1.4765	1.9085	6.923
EAS-LSDM-MoG	--	--	0.8508	0.1003	0.9802	0.951	1.831	1.0805	5.0648	1.9313	24.177
ISSMax-LSDM-MoG	20	0.769	0.8955	0.1313	0.9731	1.0269	1.8686	1.1044	4.8773	1.9999	414.781
EAS-ISSMax-LSDM-MoG	1	0.959	0.8504	0.0987	0.9801	0.9492	1.8305	1.0788	4.9516	1.9292	54.233
CSSHAD-LSDM-MoG	2	1	0.8196	0.0667	0.9716	0.8863	1.7912	1.0383	2.3451	1.8579	25.786
EAS-CSSHAD-LSDM-MoG	3	0.959	0.8502	0.0978	0.9802	0.948	1.8303	1.078	4.9317	1.9282	107.065
PTA	--	--	0.9613	0.5501	0.8883	1.5113	1.8495	1.4384	4.9229	2.3997	68.824
EAS-PTA	--	--	0.8522	0.0965	0.9799	0.9487	1.8321	1.0764	4.8027	1.9286	64.782
ISSMax-PTA	1	0.964	0.849	0.08	0.9551	0.929	1.8041	1.0351	1.7831	1.8841	144.982
EAS-ISSMax-PTA	1	0.989	0.8514	0.0968	0.9798	0.9482	1.8313	1.0766	4.8018	1.928	133.613
CSSHAD-PTA	1	1	0.849	0.0693	0.9613	0.9184	1.8103	1.0306	1.7929	1.8796	136.974
EAS-CSSHAD-PTA	1	0.997	0.8519	0.0967	0.9798	0.9486	1.8317	1.0765	4.7965	1.9284	133.07
RGAE	--	--	0.9655	0.0667	0.9887	1.0322	1.9542	1.0554	5.8771	2.0209	62.054
EAS-RGAE	--	--	0.8511	0.0555	0.9887	0.9066	1.8398	1.0442	4.8932	1.8953	57.841
ISSMax-RGAE	17	1	0.959	0.0271	0.9944	0.9861	1.9534	1.0215	4.8674	1.9805	1176.197
EAS-ISSMax-RGAE	20	0.937	0.8503	0.0553	0.9888	0.9056	1.839	1.0441	4.915	1.8944	1443.712
CSSHAD-RGAE	8	1	0.9616	0.0216	0.9961	0.9832	1.9576	1.0177	5.5125	1.9793	518.34
EAS-CSSHAD-RGAE	20	0.912	0.8506	0.0546	0.9885	0.9052	1.8391	1.0431	4.7609	1.8937	1264.956
Auto-AD	--	--	0.854	0.0163	0.9938	0.8703	1.8478	1.0101	2.6177	1.8641	88.35
EAS-Auto-AD	--	--	0.7138	0.0079	0.9925	0.7217	1.7063	1.0004	1.0629	1.7142	22.39
ISSMax-Auto-AD	20	0.294	0.879	0.0397	0.9908	0.9187	1.8699	1.0305	4.3311	1.9095	1734.455
EAS-ISSMax-Auto-AD	4	0.971	0.8736	0.0063	0.9963	0.8799	1.87	1.0026	1.7196	1.8762	201.615
CSSHAD-Auto-AD	20	0.429	0.9296	0.0474	0.9925	0.977	1.9221	1.0399	6.3265	1.9695	1771.021
EAS-CSSHAD-Auto-AD	5	0.971	0.6733	0.0101	0.991	0.6833	1.6643	1.0011	1.1175	1.6744	155.214

Auto-AD have their own best versions, EAS-CSSHAD-RXAD, EAS-ISSMax-LRR, EAS-ISSMax-LSDM-MoG, EAS-PTA, EAS-RGAE, and EAS-ISSMax-Auto-AD, respectively.

F. Discussions on Used Datasets

There are many datasets other than the four datasets used in this article have been also used in recent works. However, many of them are indeed not suitable for anomaly detection, such as San Diego airport, Gulfport, Texas coast, and Los Angeles datasets as demonstrated and investigated in [3] and [4]. The main reason is that these datasets have no complete GTs to validate anomaly detection. Also, the so-called available GTs are actually obtained by visual inspection, for example, airplanes in San Diego airport, Gulfport, Texas coast, and Los Angeles datasets, but not provided by prior

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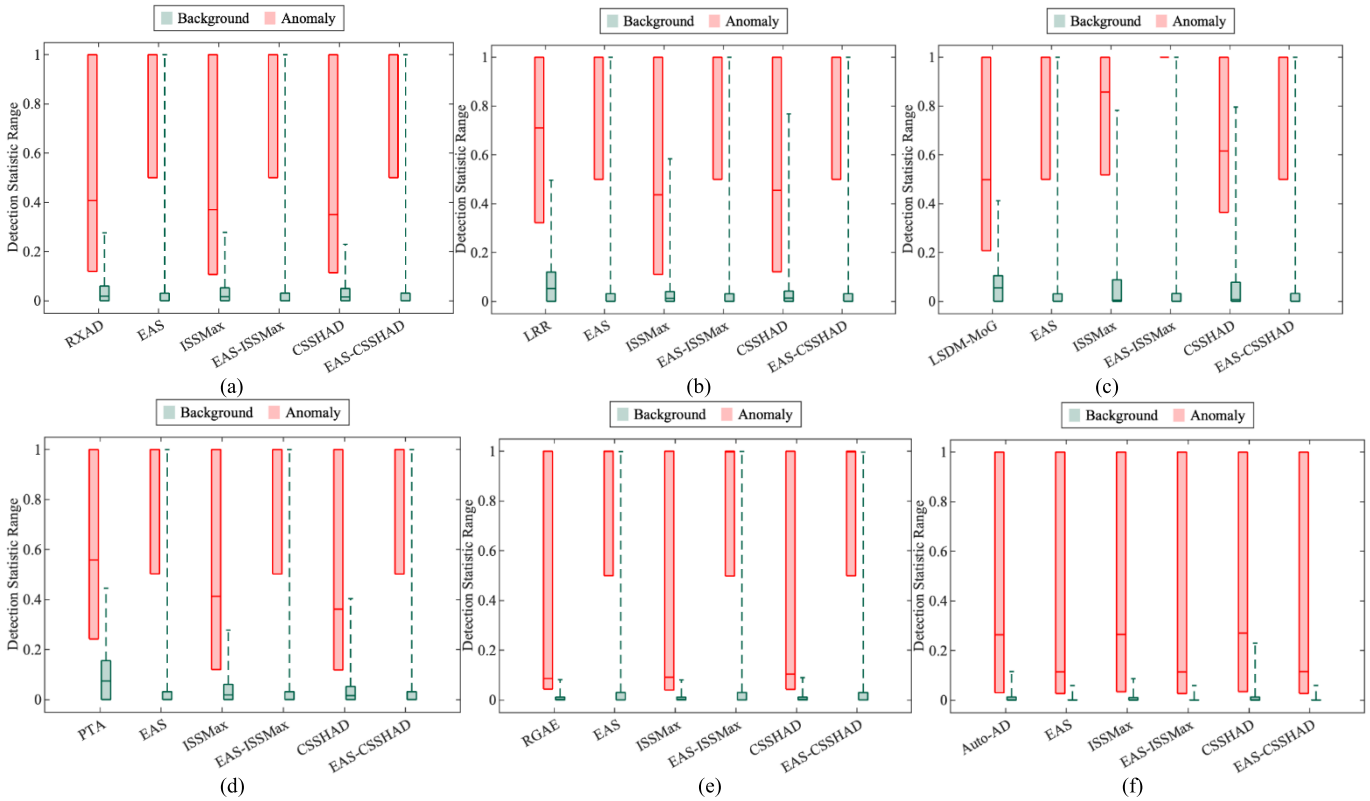


Fig. 13. BKG-anomaly SMs of (a) RXAD, (b) LRR, (c) LSDM-MoG, (d) PTA, (e) RGAE, and (f) Auto-AD, and their corresponding EAS, ISSMax, EAS-ISSMax, CSSHAD, and EAS-CSSHAD counterparts using RXAD as the anomaly detector for the AVIRIS Bay Champagne beach scene.

knowledge or ground crews. In fact, the true anomalies in these datasets are not these airplanes but rather in BKG which cannot be visually identified. In addition to these datasets, some of the datasets widely used for classification were also used for anomaly detection, such as Purdue's Indiana Indian Pines, Salinas, and Houston datasets. Unfortunately, these datasets are not appropriate for anomaly detection either. This is because the true anomalies may actually appear in their BKGs which are either not available or have been removed.

As a final remark, according to [3], most currently being used AVIRIS datasets, such as Gulfport, Texas coast, Los Angeles, including the San Diego airport scene, are not suitable for HAD. Specifically, the targets in these scenes are too large to be hardly considered as anomalies. Also, the GTs are obtained by visual inspection not prior konwledge. So, the GTs are not reliable. As a consequence, many conclusions drawn from these datasets for HAD may be misleading. The main reason that we used the San Diego airport data scene is to demonstrate its inappropriateness for anomaly detection. This scene also illustrates why all other scenes cannot be used for experiments.

VI. Further Discussions on CSSHAD

Several interesting observations and intriguing findings derived from the experiments in Sections V-B–V-E can be summarized as follows.

1) Despite the fact that ISSMax is a feedback network and CSSHAD is a feedforward network, they both performed very comparably. So did EAS-ISSMax and EAS-CSSHAD. Surprisingly, ISSMax and CSSHAD can be described respectively as an AR model which is a feedback system and an MA model which is a feedforward system in signal processing, where an AR model can be actually interpreted by an MA model [52]. With this interpretation, ISSMax and CSSHAD are indeed complement to each other and can be considered as dual HAD methods.
2) As shown in [4], EAS is designed to replace the original data space with a space which removes BKG and noise via a sandwiched model. So, all the experimental results showed that EAS could achieve significant BS, while alslo enhancing AD.
3) As shown in [3], using appropriate data scenes is critical to validate HAD methods. In particular, the San Diego data scene in Fig. 4 is more suitable for HTD than for HAD, where the three airplanes are too large to be considered as anomalous targets. This was confirmed in [3] and [4] and by experiments in Section V-D. Table VI also showed that LRR, PTA, and RGAE worked much better in detecting three airplanes than their counterparts operating in EAS and other anomaly detectors. These San Diego experiments demonstrated that LRR, PTA, and RGAE worked more like target detectors rather than anomaly detectors.
4) As for running time tabulated in Tables IV and V, CSSHAD and EAS-CSSHAD demonstrated efficient management of large datasets while lowering the chance of possible computational or data memory overhead across the four scenes. So, as an alternative to ISSHAD, CSSHAD not only performs HAD faster than ISSHAD,

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TABLE VII

AUC VALUES PRODUCED BY RXAD, LRR, LSDM-MOG, PTA, RGAE, AND AUTO-AD ALONG WITH THEIR CORRESPONDING EAS, ISSMAX, EAS-ISSMAX, CSSHAD, AND EAS-CSSHAD COUNTERPARTS USING RXAD AS THE ANOMALY DETECTOR FOR THE AVIRIS BAY CHAMPAGNE BEACH SCENE

Anomaly Detector	j	$TI_{j,j-1}$	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SBPR}	AUC_{OADP}	Time(s)
RXAD	--	--	0.9998	0.4845	0.9741	1.4844	1.9739	1.4586	18.6846	2.4584	0.068
EAS-RXAD	--	--	0.9994	0.9015	0.9901	1.9009	1.9894	1.8916	90.8681	2.891	0.054
ISSMax-RXAD	1	1	0.9998	0.4537	0.9771	1.4536	1.9769	1.4308	19.7711	2.4306	9.552
EAS-ISSMax-RXAD	1	0.981	0.9992	0.9	0.99	1.8992	1.9892	1.89	89.8643	2.8892	5.556
CSSHAD-RXAD	1	1	0.9999	0.4686	0.9783	1.4685	1.9782	1.4469	21.6423	2.4468	5.292
EAS-CSSHAD-RXAD	1	0.963	0.9994	0.9091	0.9901	1.9085	1.9895	1.8992	91.6136	2.8986	5.345
LRR	--	--	0.9997	0.6435	0.9407	1.6432	1.9405	1.5842	10.8568	2.5839	27.515
EAS-LRR	--	--	0.999	0.9	0.9903	1.899	1.9893	1.8903	92.5245	2.8893	17.766
ISSMax-LRR	1	1	0.9997	0.4847	0.9826	1.4844	1.9823	1.4673	27.9102	2.467	67.411
EAS-ISSMax-LRR	1	1	0.9992	0.9008	0.9902	1.9	1.9894	1.891	91.8421	2.8902	42.123
CSSHAD-LRR	1	1	0.9998	0.5347	0.9817	1.5345	1.9815	1.5164	29.249	2.5162	60.162
EAS-CSSHAD-LRR	1	1	0.9991	0.9	0.9902	1.8991	1.9893	1.8902	91.6762	2.8893	40.958
LSDM-MoG	--	--	0.9996	0.517	0.9387	1.5166	1.9384	1.4557	8.439	2.4553	12.458
EAS-LSDM-MoG	--	--	0.9991	0.9091	0.9902	1.9081	1.9892	1.8993	92.5651	2.8984	7.154
ISSMax-LSDM-MoG	20	0.795	0.9995	0.8116	0.98	1.8111	1.9796	1.7916	40.6543	2.7911	376.952
EAS-ISSMax-LSDM-MoG	2	1	0.9988	1	0.9901	1.9988	1.9889	1.9901	101.4919	2.9889	30.943
CSSHAD-LSDM-MoG	20	0.828	0.9997	0.6699	0.9792	1.6696	1.9789	1.6491	32.2244	2.6488	284.43
EAS-CSSHAD-LSDM-MoG	1	0.963	0.999	0.9006	0.99	1.8996	1.989	1.8906	89.7689	2.8896	18.112
PTA	--	--	0.9982	0.5371	0.9203	1.5353	1.9185	1.4574	6.7367	2.4556	64.313
EAS-PTA	--	--	0.9995	0.9012	0.9902	1.9007	1.9898	1.8914	92.3422	2.8909	69.561
ISSMax-PTA	1	0.981	0.9999	0.4794	0.9739	1.4793	1.9737	1.4533	18.3419	2.4532	154.208
EAS-ISSMax-PTA	1	0.981	0.9991	0.9009	0.9901	1.9001	1.9893	1.891	91.4096	2.8901	147.759
CSSHAD-PTA	3	1	0.9998	0.4476	0.9774	1.4475	1.9772	1.425	19.8097	2.4248	272.117
EAS-CSSHAD-PTA	1	0.963	0.999	0.9012	0.9901	1.9002	1.9891	1.8913	91.4052	2.8903	136.004
RGAE	--	--	0.9997	0.1376	0.9961	1.1373	1.9958	1.1337	35.5421	2.1334	66.192
EAS-RGAE	--	--	0.9996	0.9019	0.9905	1.9016	1.9901	1.8924	94.7595	2.892	71.547
ISSMax-RGAE	1	1	0.9998	0.1337	0.9962	1.1335	1.9959	1.1299	34.7428	2.1297	139.447
EAS-ISSMax-RGAE	2	1	0.9995	0.8992	0.9904	1.8987	1.9899	1.8896	93.5306	2.8891	207.782
CSSHAD-RGAE	1	1	0.9998	0.1429	0.996	1.1427	1.9958	1.1389	35.884	2.1387	127.051
EAS-CSSHAD-RGAE	1	0.96	0.9996	0.8981	0.9905	1.8976	1.99	1.8886	94.3058	2.8882	129.594
Auto-AD	--	--	0.9988	0.3194	0.9965	1.3181	1.9952	1.3159	90.6738	2.3147	32.291
EAS-Auto-AD	--	--	1	0.304	0.9998	1.304	1.9998	1.3038	1407.4491	2.3038	12.258
ISSMax-Auto-AD	1	1	0.9993	0.321	0.9971	1.3203	1.9964	1.3181	110.8064	2.3174	83.874
EAS-ISSMax-Auto-AD	1	1	1	0.3042	0.9998	1.3042	1.9998	1.304	1484.0732	2.304	24.451
CSSHAD-Auto-AD	1	1	0.999	0.3182	0.9966	1.3172	1.9956	1.3148	93.499	2.3138	69.579
EAS-CSSHAD-Auto-AD	1	1	1	0.3047	0.9998	1.3046	1.9998	1.3045	1471.7778	2.3045	24.414

but also maintains or enhances HAD performance. The experimental results in Section V indicated that ISSHAD which augmented the original data cube by feedback information to create a new data cube did not offer much advantage over CSSHAD which only utilized the feedforward information. In other words, each ADM in CSSHAD simply adds its output to the original data cube to construct a new data cube as an input to the next following ADM compared to each iteration in ISSHAD which uses an additional feedback process to augment a new data cube. Because of that, CSSHAD avoids creating new data cube and can reduce processing time and data volume overhead in real-world and large-scale applications.

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5) As shown in [11], ISSHAD was further extended to band sampling of EAS-ISSHAD. Using a similar approach, it is straightforward to extend CSSHAD to band sampling of EAS-CSSHAD.
6) Finally, for readers who can repeat the results presented in this article, we have made our source codes available on the following website: https://umbc.atlassian.net/wiki/spaces/rssipl/pages/27885869/10.+Download

VII. Conclusion

This article presents a feedforward network, CSSHAD, for HAD. Different from many current feedforward networks, CSSHAD does not require training samples or models to train the network. Specifically, CSSHAD implements a cascade of ADMs, one concatenated with another in sequence. Each intercascaded ADM can stand alone as an independent anomaly detector module, except that it takes the output of the preceding ADM as its input and produces its output as the input to the next following ADM. For each ADM, it performs the same anomaly detector to produce an ADMap, followed by an SF to generate a real-valued SFMap which contains spatial information about the ADMap. The last ADM can be determined by a TI-based automatic stopping rule. In order to further improve BS and enhance AD, CSSHAD can be further implemented in EAS to yield a better version of CSSHAD, EAS-CSSHAD. Finally, the frameworks of CSSHAD and EAS-CSSHAD do not specify a specific anomaly detector. Consequently, CSSHAD and EAS-CSSHAD provide a general platform for any given anomaly detector to provide better HAD performance. As a final conclusion, this article conducted comprehensive studies through extensive experiments to show that both CSSHAD and ISSHAD in [9] perform comparably in HAD. This is because ISSHAD and CSSHAD utilizes the same idea of SFMaps via a feedback process by ISSHAD and a feedforward process by CSSHAD to improve HAD performance. Accordingly, this article and [9] can be considered as companion papers.

To conclude this article, HAD currently faces challenges with few effective solutions in real-world applications. Based on many years working experience in defense applications, the goal of developing various HSI sensors is to make attempts to solve many issues that cannot be resolved by MSI sensors, specifically, in applications to anomaly detection, subpixel detection, and spectral unmixing. The first challenge is that many MSI techniques are not applicable to HSI, for instance, subpixel and mixed pixel analysis. So, developing HSI techniques must be reinvented from completely different perspectives. A second challenge is that HSI have been shifted, directed and devoted to automatic or unsupervised techniques in different applications such as reconnaissance and surveillance in defense and intelligence, environmental monitoring, food safety and inspection, pesticide pollution in agriculture, plant disease control in forestry, tumor or abnormality detection in medical imaging, and so on, all of which must be performed in a completely blind environment. A third challenge is that in the future deep space applications, HSI sensors operating on satellites should satisfy the so-called size, weight, and power (SWaP) constraints. To meet these requirements, the designed HSI techniques must be simple to be implemented and also have less computational complexity. Last but not least, HSI techniques ought to be implemented in real-time to capture timely target information. In this case, how to implement HSI techniques in real-time hardware architecture is a great challenge. Apparently, many ML and DL-based HSI techniques cannot meet the above-mentioned challenges because they all require training samples and models with a large number of parameters needed to be fine-tuned prior to testing. These techniques cannot be accomplished in real-time since the training must be updated in real-time and also requires excessive computational complexity as well as enormous time and memory for each update in real-time. The CSSHAD proposed in this article offers a feasible solution with the following advantages. The anomaly detectors used in CSSHAD can be either RXAD or a constrained energy minimization anomaly detector [30], both of which can be implemented in real-time [58]. The concatenation used in CSSHAD is a simple, direct input–output relationship without training and fine-tuned model parameters. The only extra process is to the use of an SF to produce an SFMap for each ADM. So, the challenge in implementing CSSHAD in real-time is how to design and develop a field programmable gate array (FPGA) hardware implementation, which is our ongoing research.

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